Chapter 8

Practical issues in Option Trading

Contents

8.1. Option Hedging 2
8.1.1. Hedging Using Index Futures 2
8.1.2. Option Risk Parameters 3
8.1.3. Practical hedging issues 9
8.1.4. Hedging Barrier options [Advanced Topic] 17
8.1.5. Vanna-Volga method [Advanced Topic] 21
8.2. Introduction to volatility trading [Advanced Topic] 23
8.2.1. Volatility modelling and forecasting 23
8.2.2. Trading volatility as an asset class 26
8.2.3. Trading the implied volatility surface 43
Selected references 54

Author: Ka Lok Chau

Last Revised: August 2024

8.1. Option Hedging

Hedging is an important concept in derivative pricing and trading. Before we move to the more technical discussion of these concepts, we would first want to give a general definition. When we say "the risks have been hedged," it is supposed to mean that there will be **no profit or loss** (P&L) when certain parameters move. In other words, if you manage to make money while you claim that the risks have been hedged, you must be doing a bad job! Furthermore, we need to refer to hedging against certain parameters, e.g. stock price, interest rates, volatility, and so on. A portfolio could usually not be completely hedged and there would be some residual risks. In order to make money from a position that would benefit from a certain view, one must leave the exposure to that parameter unhedged and the other risks are supposed to be hedged.

When people use the term hedging, it can refer to two different cases. Firstly, we can talk about the optimal hedging of a portfolio, and the aim is to reduce the fluctuation in P&L during times of uncertainty. What we mean is, if we are not certain of the directional movement of the underlying variables, we may want to hedge the portfolio "approximately." This strategy is often used in managing a portfolio where the aim is to achieve outperformance of the underlying basket against an index.

The other case of hedging refers to the optimal hedge of an option position. According to the pricing theory, the option price is equal to the hedging cost. We need to be able to find the correct strategy so that the option position together with the hedging instruments would result in no profit / loss when underlying parameters move. Note that for a trader managing a derivative portfolio, it is normal for the trader to adopt certain market views – the unhedged position(s) is/are the view taken by the trader.

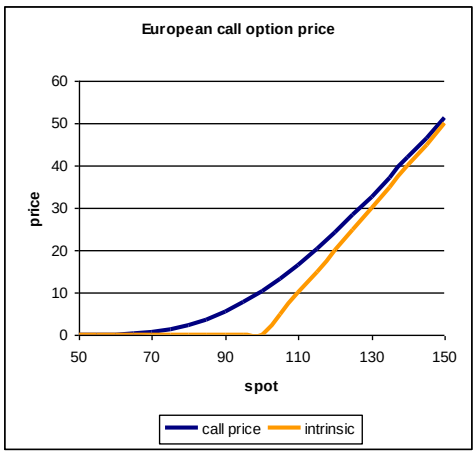
8.1.1. Hedging Using Index Futures

To hedge the risk in a (long) portfolio with futures contracts, the number of contracts that should be shorted is $\beta \frac{V_A}{V_F}$ where V_A is the value of the portfolio, β is its beta, and V_F is the value of one futures contract. Note that β is only a statistical measure; even if we hedge with this ratio, in the short term, the futures contract may not move synchronously with the portfolio which would result in fluctuations in profit or loss.

Example

Hang Seng Index futures contract is trading at 17000 and the value of a Hong Kong stock portfolio is HK$15 million, with beta of the portfolio = 1.5. The notional value of each futures contract is 17000 x $50 = $850000 ($50 being the value of 1 point). Therefore, the theoretical number of futures contracts to hedge the portfolio is: 1.5 x 15000000 / 850000 = 26.47. Since we can only trade a round number of contracts, 26 or 27 contracts would be acceptable.

8.1.2. Option Risk Parameters



The above diagram shows a price profile of a typical European call option. As we can see from the above diagram, the interesting feature for most options is that the relationship between the option price and the underlying spot price is not linear. We can calculate the sensitivities of the option price to various parameters, commonly known as option "Greeks."

8.1.2.1. Common option sensitivities for vanilla options (Greeks)

If C is the option price and S, σ, t, and r take their usual meaning, some of the common sensitivities are defined as:

First order derivatives:

Delta = Change in option price / change in share price; $\Delta = dC/dS$

Vega = Change in option price / change in implied volatility; $v = dC/d\sigma$

Theta = Change in option price / change in time to maturity; $\Theta = dC/dt$

Rho = Change in option price / change in interest rate; $\rho = dC/dr$

Second order derivative:

Gamma = Change in delta / change in share price; $\Gamma = d\Delta/dS = d^2C/dS^2$

Obviously, shares are linear instruments, and therefore only delta is non-zero (delta = 1). A share position has no gamma, vega, rho or theta, i.e. gamma=vega=rho=theta=0. On the other hand, most options are sensitive to all the other parameters. Therefore, later when we look at the issue of option hedging, delta risk can be hedged with the underlying shares, whereas other types of option risks can only be offset with instruments that are exposed to these risks, which means that one must use other options to hedge such risks of an option position.

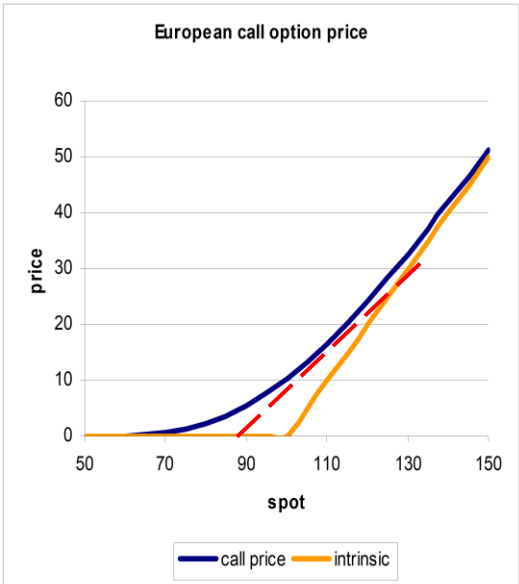
Black-Scholes Greeks

For a European call option on a non-dividend paying stock, under the Black-Scholes framework we have the following formulas:

$$\Delta = N(d_1)$$
$$\nu = S_0\sqrt{T}N'(d_1)$$
$$\Theta = -\frac{S_0N'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$$
$$\rho = KTe^{-rT}N(d_2)$$
$$\Gamma = \frac{N'(d_1)}{S_0\sigma\sqrt{T}}$$
$$\text{where } N'(x) = exp\left(-\frac{x^2}{2}\right)/(\sqrt{2\pi})$$

These results are obtained from a straightforward differentiation of the Black-Scholes formulas. Using the put-call parity, we can easily obtain the corresponding formulas for a European put.

8.1.2.2. What is option delta?



Strike 100, maturity 3 months, r=2%, volatility 50%

As defined above, delta is the sensitivity of option price to share price, or we can write $\Delta = dC/dS$ where C is the option price and S is the underlying share price. For example, if delta = 0.74, it means that if share price moves up by \$1, i.e. $dS = 1$, we can calculate $dC = 0.74 \times 1 = \$0.74$, i.e. option price would go up by \$0.74. Delta can be interpreted as the slope of the tangent (red dotted line) in the Option price–underlying price diagram.
In risk management, we often use the delta to represent the "share equivalent" of the option position (c.f. the delta-normal method in section 7.2.2). For example, if we are long 10000 options and the delta is 0.74, the position is treated as equivalent to long 7400 shares.

What is the Black-Scholes delta?

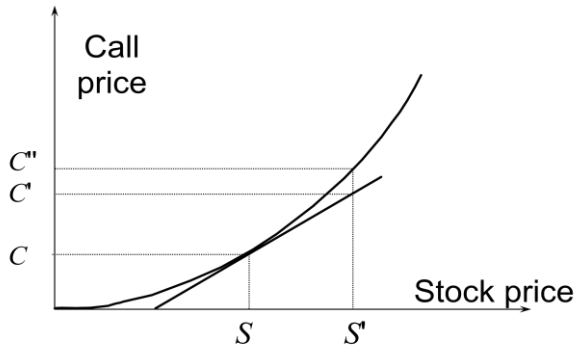
Under the Black-Scholes framework, supposedly if we keep the portfolio to be delta neutral everyday (i.e. delta of the portfolio plus any hedging instrument is always kept at zero), the final amount in our account is the same as the initial derivative price (with accumulated interest). This result is based on a few strong assumptions used in deriving the Black-Scholes formulas, such as the continuous re-balancing of the portfolio, interest rate and volatility are both constant and known. These assumptions are hard to fulfill in real life situations. For example, if we use a wrong volatility to calculate delta, we may not be able to recover the original option premium even though other conditions have been satisfied. As we have seen from the last topic, volatility is not a constant and it is impossible to predict the future volatility with accuracy.

Example of delta calculation

The delta of a European call option (Δ) for a non-dividend paying stock is: $\Delta = N(d_1)$.
If spot price = \$100, strike = \$100, time = 1, volatility = 30%, interest rate r = 3%, we could calculate Δ = 0.5987. When spot price moves to \$120 and other parameters remain constant, Δ becomes 0.8045.

Note that the delta for a vanilla option is always between -1 and 0 (for a put), 0 and 1 (for a call) (check the limits given by the yellow lines in the diagram above).

8.1.3.3. Gamma



In the above section, we use delta to represent the share equivalent position of an option. However, because of the non-linear pricing profile, the error of using delta to represent the position gets bigger if the spot price moves away from the current price. Gamma measures the hedging error caused by the curvature.

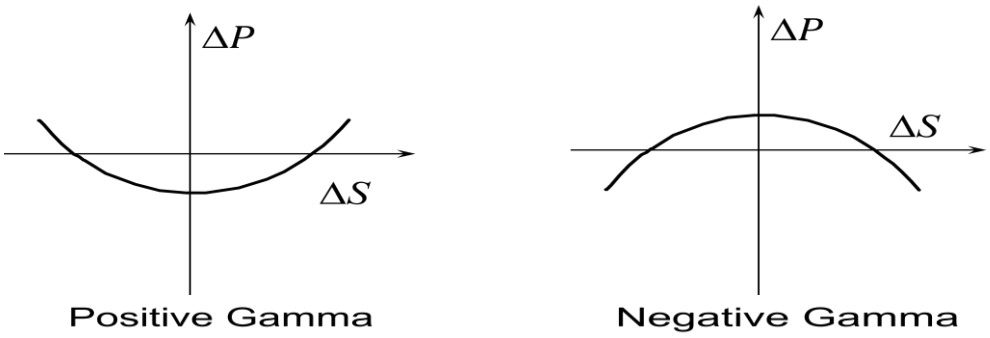
In the above diagram, when spot moves from S to S', the price change predicted by using delta alone would be C', whereas the correct option price should be C''. An adjustment term from gamma would make the approximated change much closer to the real value.

If P is the option price, a theoretical expression can be seen from a simple Taylor series expansion:

$$\Delta P = \frac{\partial P}{\partial S}\Delta S + \frac{\partial P}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 P}{\partial S^2}(\Delta S)^2 + \frac{1}{2}\frac{\partial^2 P}{\partial t^2}(\Delta t)^2 + \frac{\partial^2 P}{\partial S \partial t}\Delta S \Delta t + \ldots$$

Ignoring second order terms, this expansion links the various Greeks together, i.e.:

Change in option price = delta x dS + theta x dt + ½ gamma x $(dS)^2$

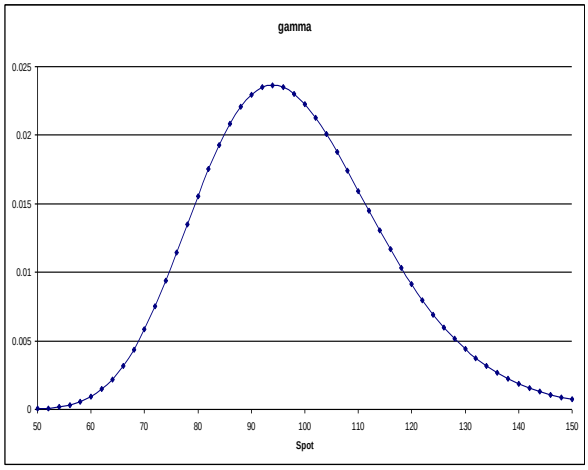


Portfolios can have positive or negative gammas, as depicted in the diagrams above. In addition, it is common to form a delta neutral portfolio, so that we can write:

$$\Delta P \approx \Theta\, \Delta t + \tfrac{1}{2}\Gamma \Delta S^2$$

Spot	Delta	Gamma
40	2.5	0.66
60	71.2	7.7
80	301.6	14.7
100	580.8	12.9
120	780.5	7.7
140	887.8	3.7
160	936.9	1.6

Long 1000 European call option
Maturity 1 year, interest rate 3%, strike 100, volatility 30%

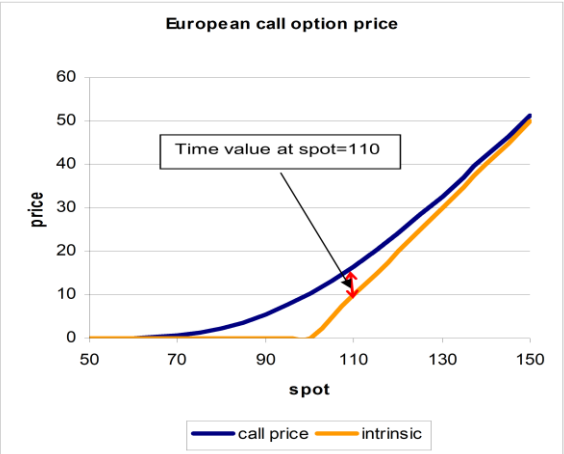


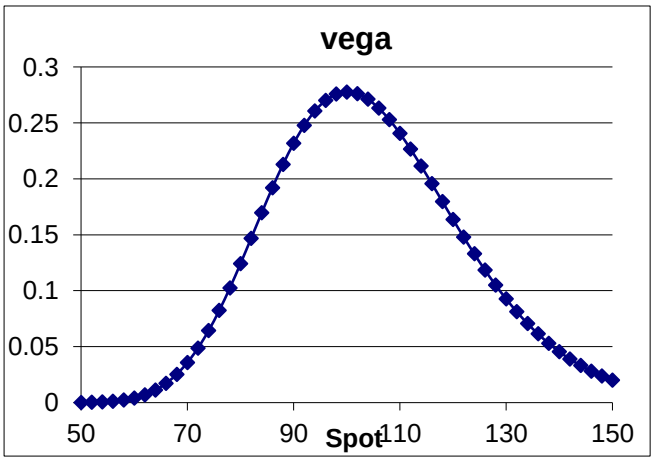
European call, Maturity 6 months, volatility 25%, strike 100, rate 3%

In the above example, when stock price goes up, delta (expressed in shares) becomes higher, i.e. the position is equivalent to holding more shares. Conversely, when stock price goes down, delta becomes smaller. We can also attribute the effect to come from a positive gamma. This is a desirable position for the trader to have, as it limits the downside risk but will fully participate in the upside. In other words, higher delta means higher gain when stock price goes up, and vice versa. Of course this position is not free, and can usually only be achieved by holding a long option position, where there is a cost to enter into this position.

Mathematically speaking, we can show that gamma is highest when spot price is near to at-the-money. We can also see this effect in the earlier option price-spot price profile, where the curvature is greatest near the to the current spot price level.

8.1.3.4. Vega, theta and rho





Option vega

Vega and theta can also be explained from the price/spot diagram for a call option. Basically, because of the non-symmetric payoff pattern, when volatility increases, it means that there is more chance of higher payoff (because spot price can move higher). However, the maximum downside remains to be 0 (when option is out-of-the-money). As the option price is the probability-weighted payoff, it is evident that higher volatility would lead to a higher theoretical option price. Similar to gamma, vega of an option is highest when the spot price is near to the strike.

Components of option theta

The blue line in the above diagram represents the option price profile before maturity, whereas the orange line is the intrinsic value (or the payoff profile at maturity). In order to explain theta, we notice that the blue line would move towards the orange line as the option approaches maturity. Given that there is less chance for the spot price to move substantially beyond the current spot price, there is a reduction in time value of the option as it moves towards maturity (c.f. section 2.3.2.). Hence the theta of a long option position is often negative, which is also known as the time decay of an option. Another explanation of a negative theta is due to the reduced possibility of the spot price moving higher because of less time to maturity, thus lowering the expected payoff.

We also need to understand the effect due to funding cost. Let's assume we have a long position of 1000 shares and the current stock price is $100. If the stock price remains at $100 tomorrow, on paper there is no P&L, but in terms of the real costs, there would be a loss of one day of interest for $100,000. The correct calculation is $100 – $100 / (1+r/365) per share where r is the annualized interest rate.

We can apply this analysis to an option example, using the following parameters: spot price = $100, strike = $50, rate = 5%, volatility = 10%, dividend = 0, European call. If the time to maturity is 365 days, the option price = $52.3810 (from Black-Scholes model). Given that the option is deep in-the-money, we can estimate its price from a cash flow analysis, i.e. calculate the forward value of $100, deduct the strike, and discount back to today, so that the option price is roughly:

= {100 x (1+0.05) – 50 } / (1+0.05) = 100 – 50/1.05 = 52.3810

One day later, the time to maturity is 364 days, and the option price = 52.3747 (from Black-Scholes model, assuming parameters remain unchanged). This is roughly given by:

= 100 – 50/(1+0.05x364/365) = 52.3747

Obviously, the change in option price (i.e. the theta) comes from the 1-day discounting effect of the strike if option is deep-in-the-money.

A peculiar situation that can happen in practice is as follows. Occasionally some option positions are already deep-in-the-money and the trader has fully hedged the positions. For example, the portfolio is long a deep-in-the-money call and short shares at 100% delta. This is a riskless position and should generate no P&L. However, in the trading system, it can show a P&L hit everyday if funding cost is not taken into account. The reason is because even if the stock price remains unchanged, the option position would show a small loss (due to negative theta) but the stock position would show no P&L.

Another look at theta

Theta can be mis-understood even by professional option traders, and one confusion comes from the difference between the settlement day and calendar day. For a trader, a common empirical definition of theta is the P&L difference between last night's closing portfolio value and the value of the portfolio before market opens today.

Let's say yesterday is Jan 28, 2014 (Tues) in Hong Kong and Jan 29 (Wed) is not a holiday. According to the discounting calculation in the section above, normally we would expect there will be one day of 'time decay' which would give rise to a negative P&L if we hold a long position in an option. In practice, a trading system would need to use the exact date schedule, i.e. cashflows would be discounted from settlement day to settlement day. On Jan 28, assuming settlement is T+2, settlement day for the position is Jan 30. On Jan 29, because Jan 31 and Feb 3 are holidays in Hong Kong, the correct settlement day is Feb 4. If we use the example above, there is a 5-day time decay between Jan 28 and 29 (because the change in the discounting effect lasts from Jan 30 to Feb 4)!

If we use the option pricing formulas correctly, theta actually comes from two components. Firstly, in calculating the forward price of the underlying, the calendar days between current date and maturity are taken into account. Secondly, for the change of discounting effect, it needs to be derived from the calendar days between settlement dates. In the example above, if the maturity date of the option is Oct 30, 2014, we will have the following day schedule:

Current date (A)	**T+2 (B)**	**Maturity (C)**	**T+2 (D)**	**days for calculating forward (C)–(A)**	**days for calculating discounting (D)–(B)**
Jan 28, 2014	Jan 30, 2014	Oct 30, 2014	Nov 3, 2014	275	277
Jan 29, 2014	Feb 4, 2014	Oct 30, 2014	Nov 3, 2014	274	272
Movement				1	5

We can see that for the first effect, a movement of one day forward would account for a 1-day theta move of the change of forward, but the second effect represents a 5-day move of the change of discounting. Under normal circumstances, the first effect would have a bigger impact on option prices. However, for options that are deep-in-the-money, the second effect becomes dominant.

Components of option rho

In order to understand rho, it is instructive to look at Black (1976) again (c.f. section 2.4.1). The European call price is given by:

$$C = exp(-rt)[FN(d_1) - KN(d_2)]$$
$$d_1 = \frac{ln(F/K) + \sigma^2 t/2}{\sigma\sqrt{t}}, d_2 = d_1 - \sigma\sqrt{t}$$

We can identify two effects of r. Firstly, forward F involves r (=$S\exp(rt)$ if interest rate is not stochastic). Secondly, r appears in the term used for discounting the expectation of the payoff, i.e. $\exp(-rt)$. If interest rate goes up, F goes higher, which means that the terms in the bracket would become bigger. At the same time, the discounting factor is

smaller, but this impact on the overall price is less. Therefore call price is more expensive, which means that the position has a positive rho.

8.1.2.5. Vanilla option characteristics and calculation

	Long call	Long put	Short call	Short put
Delta	+	−	−	+
Gamma	+	+	−	−
Vega	+	+	−	−
Theta	−	−	+	+
Rho	+	−	−	+

It would be convenient to summarize the properties of the basic options. For example, if we have a long European call position, the delta, gamma, vega and rho are all positive whereas the theta is negative. Knowing the properties of individual positions can help to understand how a portfolio behaves when more than one position is added together.

Sometimes there are no explicit formulas for the calculation of the risk parameters, whereas a model is available for calculating the option price. We can then make use of this pricing model to generate the risk parameters by first principles. For example, vega is defined as the change in option price divided by the change in volatility. We want to price a European call option with the following parameters:

Spot price = $100, strike = $100, rate = 5%, maturity = 1 year, dividend = 0
Volatility = 30%, price = 14.1730 (from Black-Scholes model)
Volatility = 31%, price = 14.5529 (from Black-Scholes model)

Therefore, the vega for this option = 14.5529 – 14.1730 = 0.3799 (approximately). In addition to the original price, only one additional call to the pricing routine is required (at volatility = 31% in this example). A slightly more accurate method is the center-difference approach which involves two calls to the pricing function. Vega is given by the difference between the prices obtained at volatility 30.5% and 29.5%, i.e. vega = Price(volatility+0.5%) – Price(volatility–0.5%). This method can also be used to obtain the other greeks.

8.1.3. Practical hedging issues

8.1.3.1. How should the portfolios be hedged?

By definition, if the portfolio is perfectly hedged, we could NOT make any money. Traders would want to enter into trades such that they could earn money when they express some market views. Therefore hedging is only needed when one is not sure of market conditions. For example, we have bought some shares and are bullish in the long run, but there may be some short term uncertainty. We may consider shorting some futures as a hedge (c.f. section 8.1.1). More often, option traders can trade volatility, i.e. they want to express a view in volatility but not the direction of the market. One way to achieve this is to hedge the directional move of the underlying share price (i.e. have a delta neutral portfolio) but leave the volatility risk (vega) unhedged.

Just to recap the definitions, a portfolio is *delta-neutral* if there is no P&L movement when spot price moves; *gamma-neutral* if delta is unchanged when spot price moves, and *vega-neutral* if there is no P&L movement when implied volatility changes. Option portfolios are rarely neutral to all these parameters. Using the example from above, if we think that the volatility would increase but are unsure whether the stock price would go up or down, we could keep the portfolio delta neutral but vega positive in order to capture some profit if our market view is correct (c.f. section 8.2.2.1).

Static and dynamic hedge

We can distinguish the cases between a static hedge and a dynamic hedge. Static hedging means that once the portfolio is hedged, there is no need for any re-balancing before the maturity of contract. This is rare unless one has identified an arbitrage strategy. It is much more common for option positions to be dynamically hedged. The hedging portfolio has to be adjusted depending on market conditions. For example, one has to buy or sell the underlying shares to hedge delta, which would create trading profit/loss. As explained in the section below, theoretically this hedging cost is equal to the option premium.

Option portfolios are often exposed to risks coming from delta, gamma and vega, among others. Obviously only instruments with gamma could hedge gamma exposures, and only instruments with vega could hedge vega exposures. Therefore, we could hedge the delta with shares, but could only hedge vega and gamma with (any) option. From the earlier sections, we also note that gamma and vega profiles could change dramatically with time and spot price. When there is a choice of different hedging instruments, one has to consider the effectiveness of the hedge, as well as other practical issues, e.g. liquidity, bid/ask spread, transaction costs (see the section on second order changes in section 8.1.4.3. below).

8.1.3.2. Issues in option hedging

Dynamic delta hedging is a key concept in the Black-Scholes framework. For example, an issuer sells 100,000 Alibaba call options. Assuming that other pricing parameters are known, when spot price = $90, delta is 55% (calculated from Black-Scholes formula). Following this result, in order to keep a delta neutral position the issuer needs to buy 55,000 shares (= 100000 x 0.55) at $90 or better when the option is traded.

On day 2, assume that the spot price has now become $93. The delta is now 61% (calculated using the new parameters). Therefore the issuer needs to buy another 6,000 shares at $93 or better. It is because the new delta is 61,000 shares but the issuer already has 55,000 shares; therefore only another 6,000 shares need to be bought.

On day 3, let's say the spot price drops to $88, and the delta is now 51% (again re-calculated from the formula). The issuer needs to sell 10,000 shares at $88 or better to maintain a delta-neutral position (61,000 – 51,000 = 10,000 shares).

In the following table, we use a similar example as above, but the issuer sold 1000 options initially and received some option premium *P*. The delta hedging operation with the share account and the balance in the bank account on a daily basis is shown below:

	Initial Position	Share Price	Delta Required	New Trade	Amount	Cash Account Balance
	(A)	(B)	(C)	(D)=(C) - (A)	(E) = – (D) x (B)	(F) = (F)previous + (E)
Day 1	0	90	550	550	-49500	-49500
Day 2	550	93	610	60	-5580	-55080
Day 3	610	88	510	-100	8800	-46280

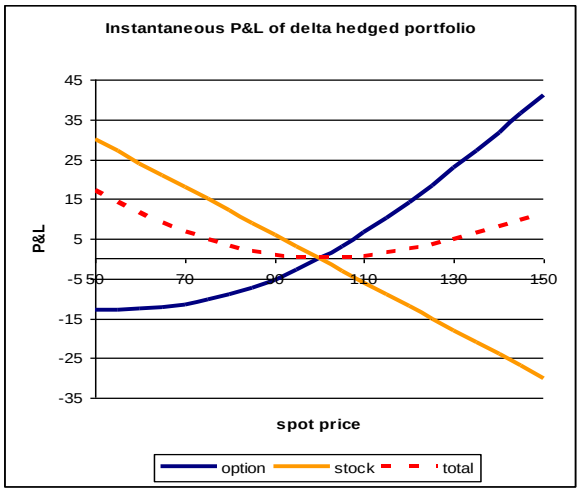
The process is repeated everyday until the option maturity. This strategy seems to lead to a certain loss, because basically we are buying shares at a high price and selling them at a low price. However, if the Black-Scholes theory is correct and the parameters are stable, *the final amount in the cash account should be equal to the initial option price, no matter whether the option is out-of-the-money or in-the-money at maturity*.

We can add three elements to make this calculation slightly more realistic. Firstly, according to the Black-Scholes framework, the hedging process should be self-financing, and interest cost should be taken in account. In the above example, we need to borrow money to buy shares. From day 1 to day 2, the change in the account balance should include the interest cost. For example, if the annual interest rate is 4%, the account balance should become –5580 – 49500x(1+0.04/365) = –55085.42 (assuming a 1-day difference in the settlement calendar). Secondly, buying and selling shares would incur transaction costs, which need to be taken away from the account balance (note that it is a *cost* no matter whether we are buying or selling shares). Thirdly, one might need to add back the effects due to dividends. In the case of a short position, the accrued dividends should be paid to the stock lender, so this amount has to be deducted from the account balance.

In practice, the Black-Scholes hedging scheme cannot work so perfectly. We already include transaction costs in the calculations above, but other factors which may affect the accuracy of the hedging operation include:

- Wrong estimate of volatility;
- Intra-day movement (notice that volatility is usually measured based on closing prices);
- Gap movement in underlying price;
- Liquidity constraints, i.e. it may not be possible to execute a big trading order without affecting the stock price.

P&L profile of delta-hedged position



The example in the previous paragraph looks at the effect of the P&L when the hedging action is continued everyday. Now we examine the hedging issue from a different angle. Assume the current spot price is $100 and we have a portfolio with long 1 European call option and short Δ shares, so that the portfolio is delta-neutral. This diagram shows the instantaneous P&L of this portfolio when underlying spot price moves, where the aggregate profile is shown as the red dotted line. Because the price profile (the blue line) is a convex function of the spot price whereas the hedge (the orange line) is a straight line, if stock price has suddenly moved away from $100, we would always make a profit! The profit increases if the stock price movement is bigger. This seems too good to be true, but of course there is no free lunch – *if stock price remains unchanged tomorrow, there would be a loss on time value for the option position* (i.e. maturity is reduced everyday).

P&L attribution

Instrument	position	stock price ($)	volatility	MTM price	delta (shares)	vega ($)	P&L ($)
Day 1							
Call option	10000	100	25%	11.344	5788	3878	0
shares	–8000	100		100	–8000	0	0
total					–2212	3878	0
Day 2							
call option	10000	105	26%	14.894	6527	3797	35500
shares	–8000	105			–8000		–40000
total					–1473	3797	–4500

In this example, the portfolio has long 10000 call options and short 8000 shares as hedge. The portfolio is not delta or vega neutral: initially we are short delta (–2212) and long vega ($3878). When stock price and volatility both go up, the change in P&L can be positive or negative, depending on whether the delta effect or the vega effect is more dominant. Assume that the stock price has increased by $5 and the volatility also moves up by 1% after 1 day. When delta and vega of the portfolio are available, we can make use of these parameters to estimate the P&L change when the parameters move (without using a pricing model to re-price all the positions). A rough estimate of the P&L is given as follows:
P&L from delta = –1800 x 5 = –$9000; –1800 is roughly the average of –2212 and –1473
P&L from vega = $3800 (approximately), given that the volatility has moved up 1%.
total = –9000+3800 = –$5200 (compared to the actual P&L of –$4500 from the pricing model).

8.1.3.3. Hedging strategies

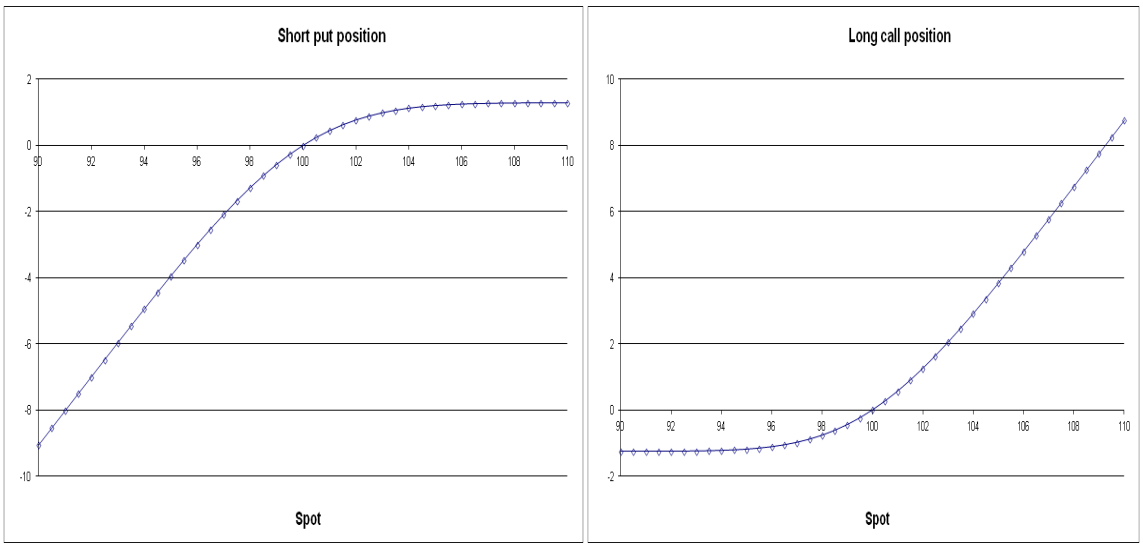
While the above sections try to analyze the risks of individual option positions, it is rare to hedge a single position. In practice, risks of a portfolio with many option positions would be managed together. The trader would need to keep within trading limits (typically with limits in delta, gamma, vega, theta). Typically, a trader would apply some kind of hedge to the first order price risk (i.e. delta), and trade the risks that come from the change in delta or adopt volatility trading strategies. Furthermore, unlike the case for demonstrating the Black-Scholes framework above, a hedge is only applied for short term and seldom

throughout the life of an option, because supposedly the option position can be closed at any time.

A main difficulty in option risk management is the problem of aggregating the risks. Even if we consider the trading of a stock option portfolio with stocks from a single country, usually there are two levels of aggregation. Firstly, the aggregation can be performed based on each single underlying. For example, there can be many different options on a stock index with different strikes and maturities. The risks of options with all these different characteristics are simply added together, to form the total delta, gamma, vega, theta for this underlying. This is a crude method because risk parameters can change rapidly and changes in the market parameters can affect options with different characteristics in uneven ways (see an example in the section on second order changes below).

The second level of aggregation is to calculate the summary risk parameters across all underlyings, normally on a per country basis. For example, we can have a delta of $2 million (using a dollar amount instead of number of shares to represent the delta) for the Hong Kong portfolio. This is supposed to mean that when the market moves up 1%, the profit of this portfolio is roughly $20,000. Of course, this is only an approximation at best, since the inherent assumption is that the same shock (% move) is applied to all the underlyings. In reality, stocks would have different movements everyday. However, aggregate risks on a country level can be useful if a trading desk has a portfolio with positions in many different countries.

Using delta to represent position risk

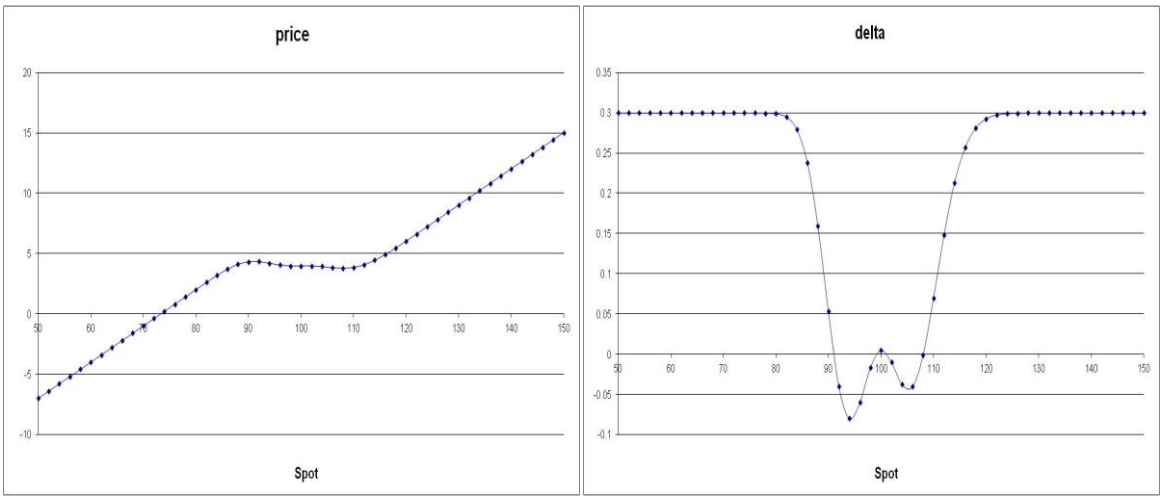


One must be beware of reading the risk parameters directly and not understand how they can move when underlying conditions change (especially if second order parameters such as gamma are included). In both diagrams above, the delta at the current spot price (assume = $100) is the same; it is obvious that their risk profiles are very different. One method to deal with this problem is to separate between upside and downside risks, but it means that more calculations have to be performed to generate the risk parameters and aggregating them can still be a challenge.

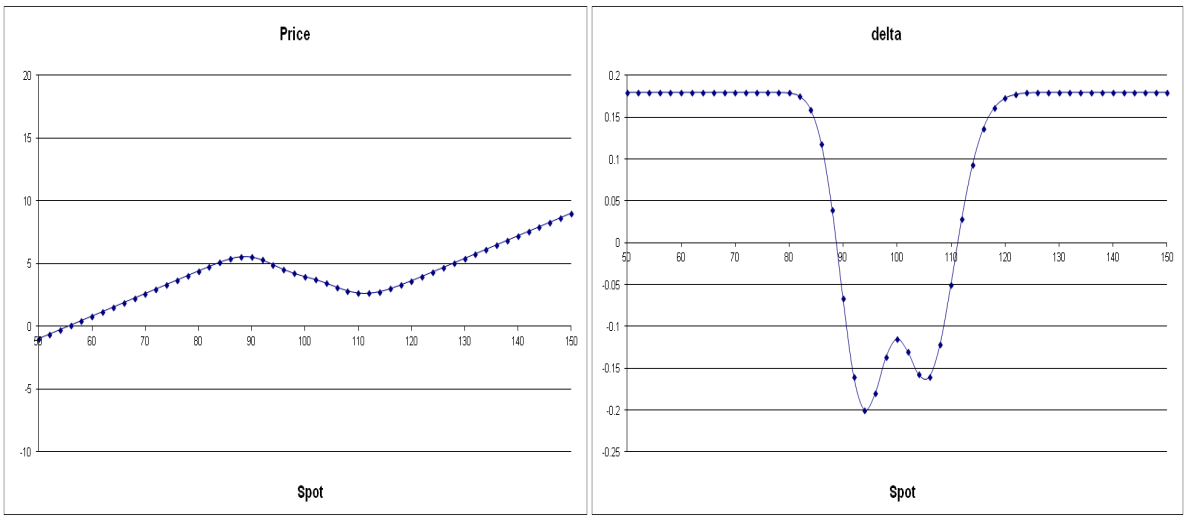
Delta hedging example

Traders may not always want to keep the portfolio exactly delta neutral (even if they don't want to express a view in the underlying change). Consider the following positions:

short a 92 strike put, long a 96 strike call, short a 104 strike call, and long a 108 strike call. Current spot price = \$100, r=1%, σ=16%, T=0.083 (1 month); delta of this portfolio = 0.7.



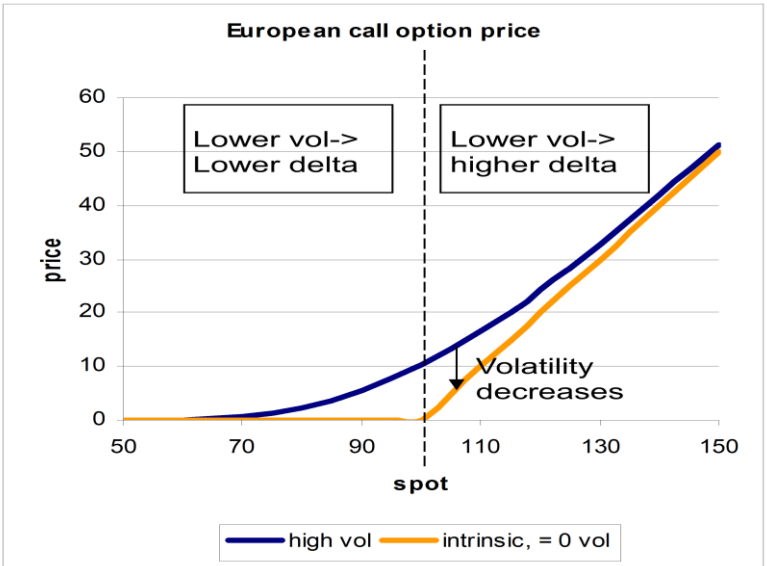
If we hedge the portfolio to make it delta neutral by shorting 0.7 share, the resulting price and delta profiles are shown in the above diagrams. It is evident that when the spot price lies between \$90 and \$100, the hedge is pretty effective and the change in price of the resulting portfolio is quite small. However, for big movements in the stock prices, the portfolio resembles a long stock position (with a constant delta of 0.3 for spot below \$80 and above \$120). There can be big variations in the portfolio prices.



Sometimes traders would adopt a modified delta hedge. Instead of hedging with a delta of –0.7, let's assume the hedge applied is –0.82. It can be seen that while there may be some residual P&L around the current spot price, the overall P&L is limited across a wider range of spot price movement; this is useful especially when the spot price is volatile.

Volatility and delta

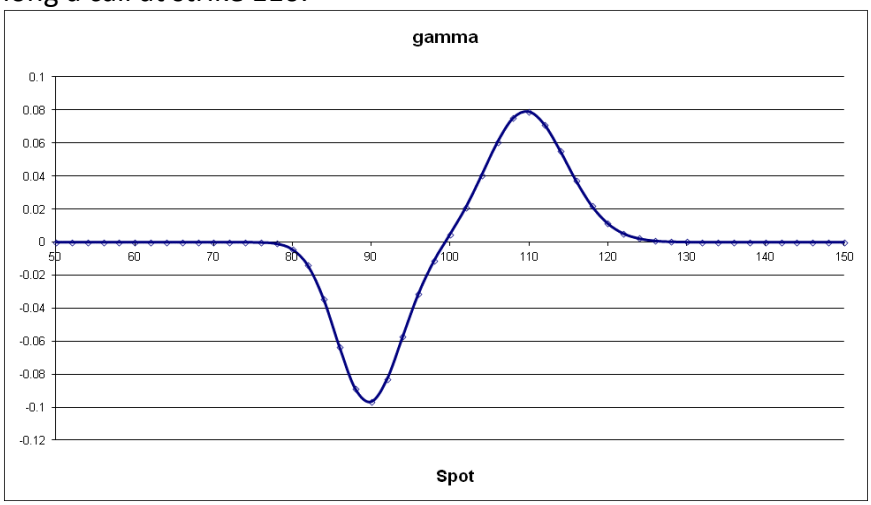
In real market conditions, volatility is not a constant and sometimes we need to change the volatility that we use to price the options. If the portfolio is initially delta-neutral and we change the volatility marking, how would delta move?



The answer depends on whether the option is in-the-money or out-of-the-money. We know that as volatility decreases, the option profile would move towards the intrinsic value, i.e. the blue line would move towards the orange line if volatility becomes zero. Remember that delta is the slope of the tangent of the option payoff. In the case of a call option, if the option is in-the-money (i.e. the right hand side of the dotted line), the slope of the tangent would increase when the blue line moves towards the yellow line (from a positive number smaller than one and move towards one). On the other hand, when the option is out-of-the-money, the slope of the tangent would decrease (from a positive number smaller than one and move towards zero).

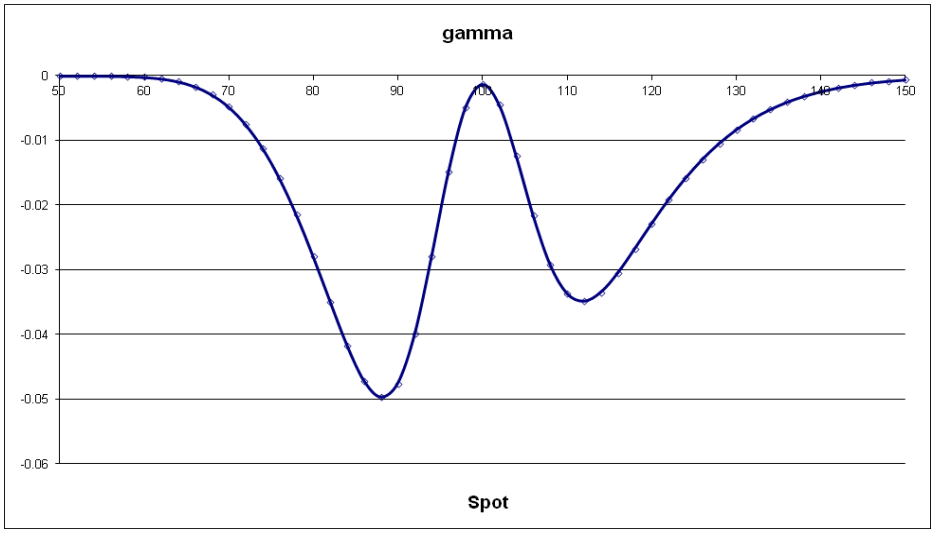
Gamma for a portfolio

We illustrate the potential problem of relying on the gamma alone to assess the characteristics of a portfolio. In the first example, we have a portfolio which is short a put at strike 90 and long a call at strike 110.



The gamma is almost 0 at the current spot price (=100), but changes rapidly (and at different signs) when spot price moves away from the current level.

In the second example, we have a long position in a ratio calendar spread: long 1 call at strike 100, maturity 1 month; short 2.5 calls at strike 100, maturity 6 months.



Again, the gamma is close to 0 at the current spot level (=100), but significant gamma persists at different spot price levels. These examples show that, even if the gamma of a portfolio is shown to be zero, there may be third order risks which cannot be identified by the gamma measure alone.

Second order changes

Assume the spot price of the underlying stock is $100, and we have the following position:

- European call, strike $100, maturity 1 year, volatility 25%, rate 4%; vega = 0.3831.

We would like to hedge so that the portfolio is vega neutral, and the following instruments are available:

- Option *A* – European call, strike $100, maturity 1 month, volatility 30%, rate 4%; vega = 0.1148.
- Option *B* – European call, strike $100, maturity 2 years, volatility 23%, rate 4%; vega = 0.5194.

All these options are based on the same underlying stock. In order to match the vega of the original position, we could hedge with either 0.3831 / 0.1148 = 3.337 of option *A*, or 0.3831 / 0.5194 = 0.7376 of option *B*. The question is, which one should we choose as a better hedge?

From the trading point of view, option *A* is usually more liquid (because of the shorter maturity) and is preferable if we only need to consider the ease of entering and exiting the position. However, another consideration is the effectiveness of the hedge when market parameters move. The simplest case is if spot price stays at $100, and both *A* and *B* seem to work fine. However, if spot price moves to $110, the vegas would become (calculated from the Black-Scholes answer and based on the new parameters):

Vega of original position = 0.3519;
Vega of *A* = 0.0400; net vega = 0.3519 – 3.337x0.04 = 0.2184;
Vega of *B* = 0.4869; net vega = 0.3519 – 0.7376x0.4869 = –0.0072.

Therefore, we can see that if we hedge with option *A*, there will be a big residual vega if underlying spot price moves away from the current level. So if both the spot price and volatility moves, say the volatility moves down while the spot is at $110, hedging with option *A* would suffer a bigger loss than hedging with option *B*. The lesson is that the Greeks could change when underlying market parameters change at the same time. Hence it is advisable to perform some scenario analysis to check whether the hedge parameters change much when certain other parameters move.

The risk parameters only represent a snapshot of the risks at the current parameters. Even if the risks are well hedged, there may be higher moment exposures and there may not be enough time to adjust the hedges (gap moves). We already show an example of a modified hedge above. For managing the risk of complex portfolios, different scenario reports have to be generated at least on a daily basis. For example, we can shift forward the time to maturity (i.e. reduce the time to maturities of all positions), shift the underlying spot price levels and the volatilities of the underlyings, and observe the change in the risk parameters accordingly. This would help to reduce the "surprise factor" when unpredictable market conditions occur, and the trader would be able to have a better understanding of the characteristics of the portfolio.

Vega for a portfolio

Different maturities have different volatility shifts as the volatility surface does not always move in parallel (although some research shows that parallel shifts account for more than 90% of the movement of the surface (see Kaye (2013)). For risk management purposes, vega is broken down into maturity buckets and a weighted shift of vega is calculated. One possible method is to use a volatility forecasting model like GARCH to provide estimates of volatility movements according to the different maturity buckets.

When we have to calculate an aggregate vega for a portfolio, we need to understand what kind of assumptions are made. As a start, a portfolio will often include many underlyings. Simply adding the numeric value of vega basically says that the volatilities of all the underlyings will move with the same magnitude, which obviously would never be the case.

8.1.4. Hedging Barrier options [Advanced Topic]

Barrier option is one of the earliest types of exotic option traded in the market, and it can be found in many structured products because of the cheaper costs. The main feature of a barrier option is that when the spot price hits a barrier, the option is terminated ("knock out"). There is also a "knock-in" option where the option only exists when the barrier is touched during its life. These options are cheaper than the vanilla calls and puts because of lower chance of payoffs.

From the risk management point of view, barrier options can be divided into two types. In our first example, a European call option has a strike = 100, barrier = 95, and the current spot price is 100. This is an example of a standard down-and-out call option. The

option is out-of-the-money when it gets knock out, and hence the risk is comparatively lower.

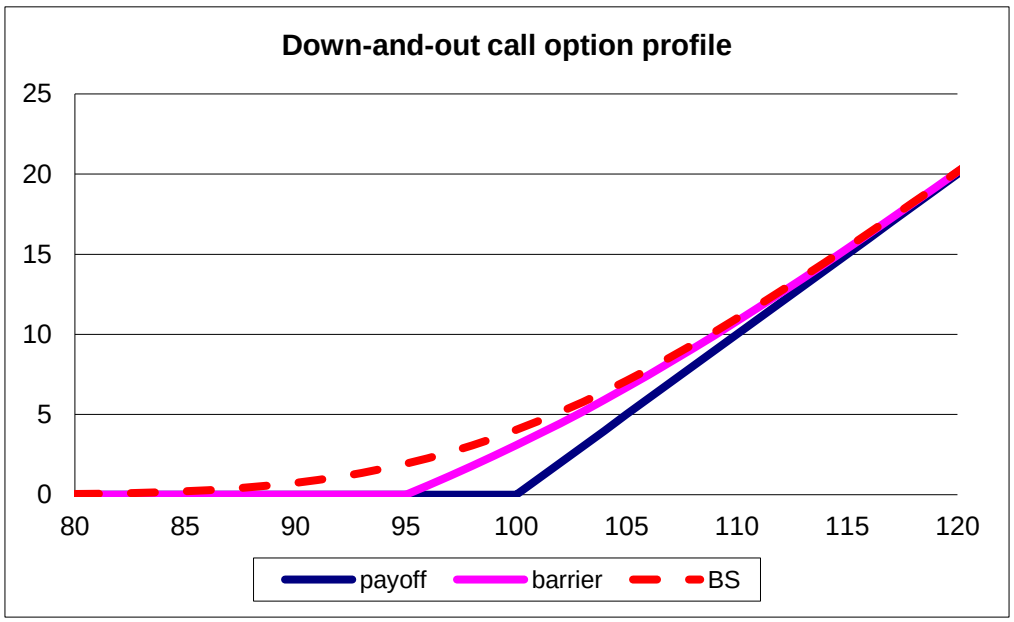
In another variant, the option can knock out when it is in-the-money, e.g. European call option with strike = 100, barrier = 130, current spot price is 100. This is risky as the payoff can drop suddenly when the barrier is hit. Imagine the situation when the option has only got one more day to maturity, and the spot price is at 129. If the price stays at this level, the payout at maturity is 29. However, if spot price goes up by 1 dollar, the payout becomes 0. However, not all in-the-money barrier options are risky. One example is the Callable Bull Bear Contract in Hong Kong (course notes section 3.2). Assume a European call option where the current spot price is 22400, strike is 21800, barrier is 22000. We can see that it is an in-the-money barrier option (i.e. when the barrier is hit, the option has a certain payout), the payoff profile is smooth and has no discontinuity.

Depending on the different markets, barriers can be monitored continuously or discretely. If it is a continuous barrier, it means that we need to be able to observe the underlying asset price at tick level, hence a liquid underlying is required. Normally it is not suitable for baskets of underlyings (e.g. a stock basket) and continuous barriers are usually found only in the FX market.

Many instruments have discrete barriers (e.g. monitored daily or even quarterly). This is more suitable for baskets and equity underlyings. Theoretically, monitoring less frequently would mean that the price of a knock-out option would increase (because there is less chance of knocking out and paying 0). If we only have pricing models for continuous barriers, we need to apply an adjustment to the fair price e.g. see Broadie, Glasserman, and Kou (1997). In general, note that the pricing of barrier options would be much affected by the volatility at the barrier, hence a model which takes into account of volatility skew is essential.

8.1.4.1. Down-and-out call

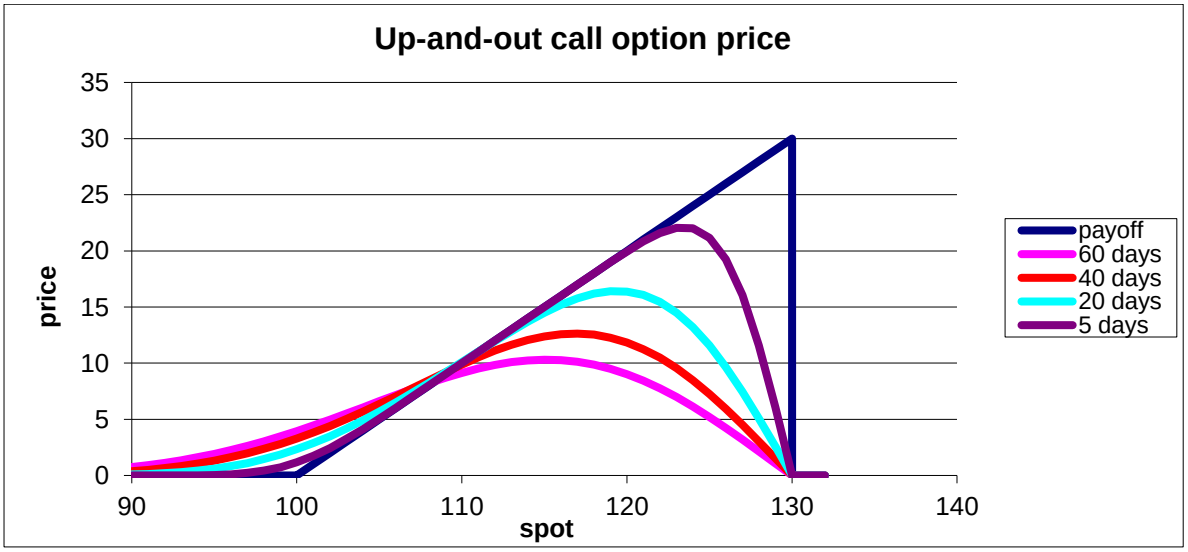
- Strike 100, Barrier 95, volatility at strike 25%, volatility at barrier 27%, interest rate = 0, maturity 60 days



The above diagram shows the price profile of a down-and-out barrier call option. While the delta is discontinuous at the barrier (changes from a positive number to 0), the absolute delta is small because the option is out-of-the-money at that spot level, hence the risk is not substantial.

8.1.4.2. Up-and-out call

- Strike 100, Barrier 130, volatility at strike 25%, volatility at barrier 22%, interest rate = 0

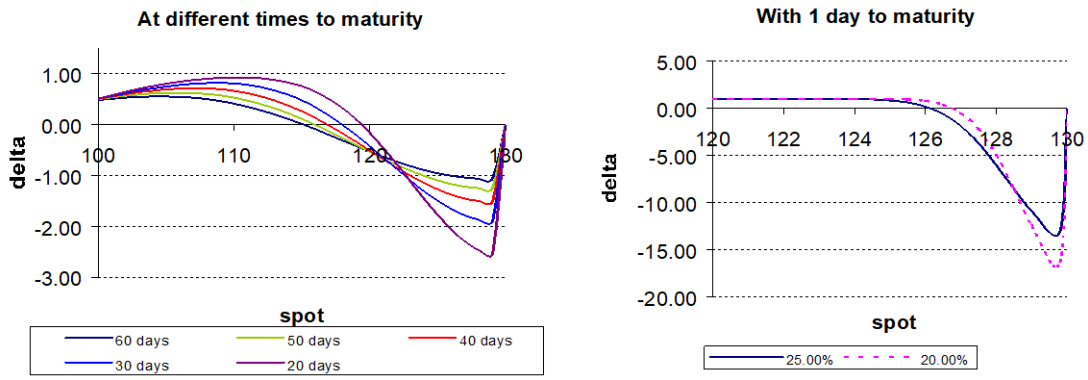


The risks of up-and-out-call options are very different from the vanilla options. The lines represent the profiles at various times to maturity. We can see a number of interesting features from the diagram above. We note that both gamma and theta change signs in the different regions of spot prices. In this example, when spot price is above 110, theta is positive – as time to maturity is reduced, the option price becomes higher while spot price is kept constant. However, when spot price is below 110, theta is negative. Another important feature is that gamma will also vary according to spot price level. As an example, with 20 days to maturity, the gamma turns from positive to negative when spot price goes from around 115 to 120. Right at the barrier level of 130, the gamma becomes positive again (as delta increases, with a discontinuity). All these features pose serious challenges if a trader needs to hedge this position.

Up-and-out call delta profile

- Strike 100, Barrier 130, volatility at strike 25%, volatility at barrier 22%, interest rate = 0

Figure 5. Delta of up-and-out call option



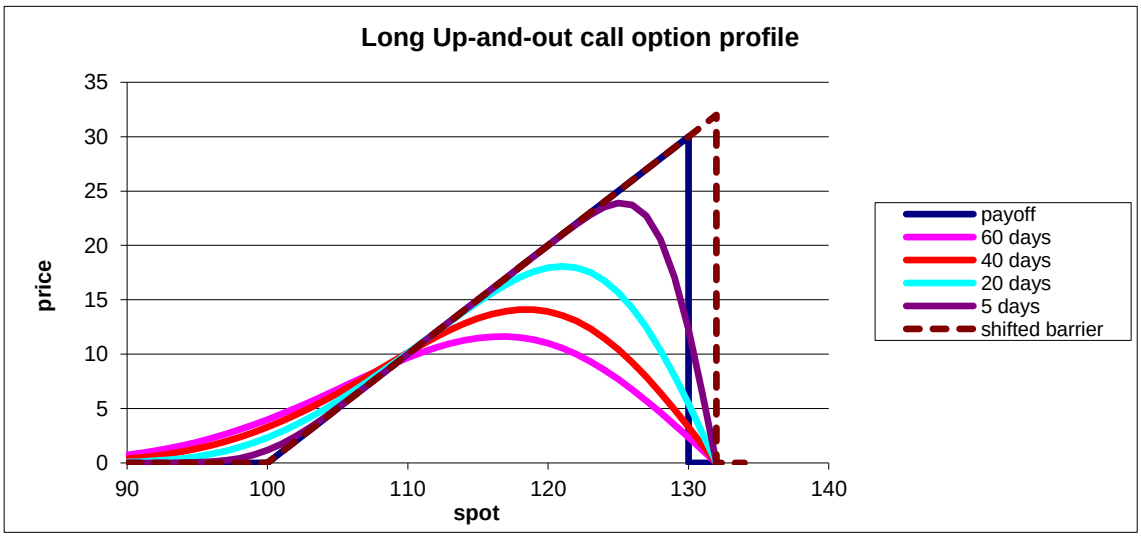
We look at the delta hedging in more detail. In this example, with 60 days to maturity, the delta turns negative when spot is at about 115 (negative gamma position starts when spot is slightly higher than 100). Note that delta could become very high near maturity when the spot is close to the barrier (more than 15 times the notional in the right diagram). In fact, the delta is higher when the barrier volatility is lower; this is because of the bigger discontinuity as seen at the barrier.

Traders are used to handle both long and short option positions, with positive or negative gamma. We have seen above that a long position in an up-and-out call will generate a big short gamma position when spot price is at some intermediate spot level, and the opposite is true for a short position. However, given that negative gamma positions are more difficult to manage, it is particular undesirable to have big negative gamma positions. Hence investment banks (IBs) would only want to short an up-and-out call.

With a short position, it is true that the gamma would be highly negative near to the barrier level, but since hitting the barrier is a one-off event, the risk can be managed through a trading strategy. A particularly popular and intuitive risk management technique is to book the trade with a shifted barrier. Remember that our objective is to try to reduce the maximum delta in order to mitigate the gamma risk. In terms of pricing, the price from a shifted barrier may not be competitive if the barrier is shifted too far. However, it is an effective way of handling the risks, and by trading the delta in an appropriate manner (buying back, and pushing the level, as described below) there is a potential for a substantial profit from the hedging action.

A shifted barrier means that although we know that the real barrier of the trade is X, we book the trade in the system with a barrier set to Y. In the example below, the barrier is 130, whereas the profiles for a barrier at 132 are shown.

- Strike 100, Barrier 132, volatility at strike 25%, volatility at barrier 22%, interest rate = 0



The IB has a short up-and-out call position, thus the hedge is equivalent to a long position (same as the diagram above), i.e. would have a short stock position near to the barrier. For effective hedging, we want to maintain a delta of 0 once the barrier is breached. Since the hedge has a big negative delta near to the barrier, we need to start to buy back the delta when spot price is approaching 130. By shifting the barrier to 132, we can see that the (negative) delta is still quite high near to the "real" barrier level of 130. Since the delta is

high (more than 100%), it is possible that the spot price can be pushed higher because of the big buying order.

Once the barrier is breached, the option is knocked out and the trader can realize a "windfall" mark-to-market gain, because in the option book, the option is deemed to be alive and has a negative value (because of the shifted barrier).

8.1.5. Vanna-Volga method [Advanced Topic]

Various models have been developed for the pricing and hedging of derivative contracts. The inherent link between the two procedures is that the fair price of a contract should reflect its hedging costs. Even in the absence of a sophisticated model, market practitioners have used some kind of interpolation procedures in determining the prices of exotic derivatives. In the Black-Scholes framework, it is assumed that the volatility is constant, whereas in reality the implied volatility moves frequently. An ad-hoc procedure for taking into account the impact of volatility smile into the pricing of first generation exotic options, known as Vanna-Volga method (VV method, or the "Traders' Rule of Thumb"), is very popular, especially in the currency market.[1]

When implied volatility moves, the change in vega ν of a position is given by:

$$\delta v = \frac{\partial v}{\partial S}\delta S + \frac{\partial v}{\partial \sigma}\delta\sigma$$

where Vanna is defined as:

$$\frac{\partial v}{\partial S} = -\frac{v d_2(K)}{S_0 \sigma \sqrt{T}}$$

and Volga is defined as:

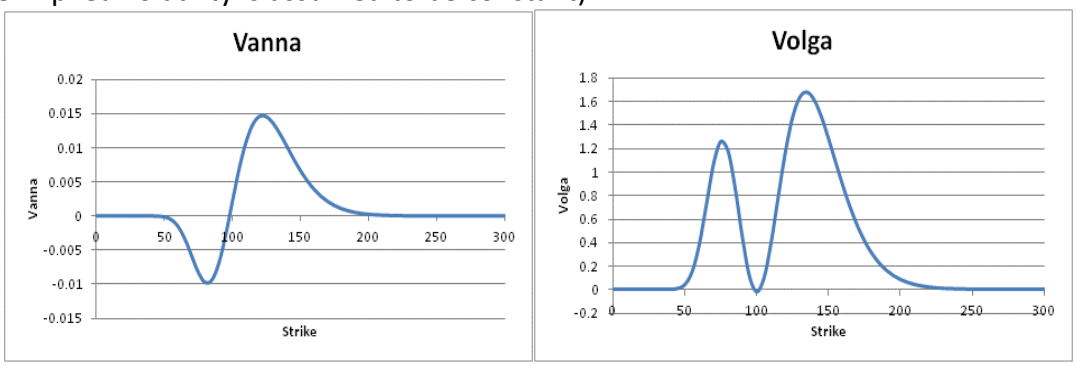
$$\frac{\partial v}{\partial \sigma} = \frac{v d_1(K) d_2(K)}{\sigma}$$

These expressions refer to the standard Black-Scholes framework, where:

$$d_1(K) = [ln(F/K) + \sigma^2 T/2]/\sigma\sqrt{T},\ d_2(K) = d_1(K) - \sigma\sqrt{T}$$

Note that in the definition for Volga above, changes in volatility refer to a parallel shift of the whole volatility surface.

With the following parameters: Forward = 100, interest rate = 0, implied volatility = 20%, time = 1 year, the vanna and Volga profiles with varying strikes are shown below (note that the implied volatility is assumed to be constant):



[1] For example, refer to Antionio Castagno (2010) and Uwe Wystup (2006). A good and concise summary can be found in Bossens, Rayee, Skantzos and Deelstra (2010).

Pricing with Vanna and Volga

In order to hedge an option with volatility changes, FX traders will perform:

- a delta hedge through spot FX
- a vega hedge through an ATM straddle
- vanna hedged through a risk reversal (RR)
- volga hedged with a butterfly spread (BF)

The meanings of RR and BF are defined as:

$$RR(\Delta) = call(\Delta) - put(\Delta)$$
$$BF(\Delta) = \frac{[call(\Delta) + put(\Delta)]}{2} - \frac{[call(\Delta_0) + put(\Delta_0)]}{2}$$

where $call(\Delta)$ refers to a call option where its delta = Δ (typically this is set to 0.25) and $call(\Delta_0)$ refers to an ATM call; for a put, $put(\Delta)$ is an option with delta = $-\Delta$. From the Vanna and Volga profiles, we can see that the *RR* is almost like a pure Vanna instrument (with little Volga), whereas the *BF* is the opposite: an almost pure Volga instrument with little Vanna. We can see these from a simple example. Assume that when Δ = 0.25, the strike of the call is 125 and the strike of the put is 75. The vanna of a risk reversal is the difference between the vanna of the call and the put, which is equal to roughly +0.025 from the diagram above. On the other hand, the Volga is more of less the difference between the two peak, resulting in a very small value. Using a similar reasoning, we can see that the butterfly strategy has a very small vanna, but a much higher Volga.

The pricing of a derivative contract with the VV method is given by the following equation:

$$\begin{aligned} X^{VV} &= X^0 + [\text{cost of Vanna} + \text{cost of Volga}] \\ &= X^0 + \alpha[RR(\Delta) - RR^0(\Delta)] + \beta[BF(\Delta) - BF^0(\Delta)] \end{aligned}$$

where the superscript zero denotes pricing the derivative with flat ATM volatility, and the coefficients α and β are given by:

$$\alpha = \frac{\text{Vanna}(X)}{\text{Vanna}(RR^0)},\ \beta = \frac{\text{Volga}(X)}{\text{Volga}(BF^0)}$$

This method has been used to price first generation exotics like barrier option (FX knockout) or double barrier option (double no-touch). However, the prices obtained through this method are often found to be too high, leading to a heuristic adjustment as follows:

$$X^{VV} = X^0 + (1-p)[\text{cost of Vanna} + \text{cost of Volga}]$$

One explanation of this adjustment is the possibility of a jump in the underlying spot price leading to an over-hedge where the hedge is not required. p is seen as the probability of jumps in the underlying, causing the spot to move through the barrier (a market for p exists for different types of options). However, this is more of an excuse than a real explanation.

The method is commonly used in the FX market, probably because of the idiosyncratic market conventions. Typically the prices of RR and BF are quoted instead of options at individual strikes, and hence it will be convenient to make use of these market quotes to arrive at the fair prices of exotic options, whereas the hedges can also become explicit. An application of the method is that we can use it to price vanilla options and infer

the volatility smile from three fixed points. The results show that they closely resemble the market quotes at various strikes.

Compare with more sophisticated methods like local volatility or stochastic volatility models, the Vanna-Volga method is more intuitive to traders, but may underestimate the inherent risks of complex derivatives. From a modeling point of view, it may not be arbitrage free (c.f. local volatility models). Given that this is only a heuristic method, one must be careful to show arbitrage bounds (e.g. see Castagna and Mercurio (2007)).

8.2. Introduction to volatility trading [Advanced Topic][2]

8.2.1. Volatility modelling and forecasting

The standard deviation of return, also known as volatility in option pricing context, is a key input parameter in many important financial models. Risk management applications (such as VaR), asset allocation (portfolio optimization), and option pricing cannot work without an accurate forecast of this parameter. However, volatility is not constant and cannot easily be observed. While it is possible to calculate its value from a historical time series (as discussed in section 2.4.2), the actual requirement is the future value instead.

Two different methods have been used to estimate this value:

- If option prices of the underlying are already available, it is possible to arrive at the implied volatility, noting that: market option price $P = BS(S,K,r,d,t,\sigma)$, where $BS()$ is the Black-Scholes pricing function. Given that P,S,K,r,d,t are all known, σ can be backed out by a simple root-searching algorithm as the $BS()$ function is non-linear.
- Using econometric methods, historical data can be utilized to generate a forecast. However, there is no guarantee that this forecast will be realized.

8.2.1.1. ARCH and GARCH

The autoregressive conditional heteroscedastic model (ARCH), introduced in Engle (1982), is the starting point of understanding the more sophisticated models. A popular extension is GARCH, which stands for Generalized Autoregressive Conditional Heteroscedasticity (Bollerslev (1986)). Volatility would change in the future depending on its current level, with two characteristics:

- Long term volatility tends to revert to a constant level (mean reverting behavior).
- More weights are assigned to more recent observations in deriving the model parameters.

Other extensions such as Exponential GARCH (or EGARCH, Nelson (1991)) have also been proposed, but they are not as widely accepted as GARCH.

ARCH model

Starting from the AR(1) model,[3] the process can be described by:

$$r_t = \phi_0 + \phi_1 r_{t-1} + a_t$$

[2] Some sections in this part have not been updated with the latest data.

[3] c.f. Course notes Appendix 2 on an introduction to financial time series analysis.

where a_t comes from a random process with mean 0 and constant variance σ^2. As extra information is available (in the form of the value of r_{t-1}), the conditional mean of r_t, given the information at $t-1$, would be

$$E(r_t|r_{t-1}) = E(\phi_0 + \phi_1 r_{t-1} + a_t) = \phi_0 + \phi_1 r_{t-1}$$

Note that this value is different from the long run expected mean of r_t, which would be $E(r_t) = \mu = \phi_0 / (1-\phi_1)$. Usually the conditional mean gives a better prediction of the value of r_t compared with the long run expected mean. We can also write down the conditional variance of r_t, given the information at $t-1$:

$$\mathrm{Var}(r_t|r_{t-1}) = \mathrm{Var}(a_t) = \sigma^2$$

In the AR model, the variance of the white noise process a_t is constant. The ARCH model allows the conditional variance of r_t ($=\sigma^2$) to also evolve with an AR(m) process, such that

$$\sigma_t^2 = \alpha_0 + \alpha_1 \sigma_{t-1}^2 + \cdots + \alpha_m \sigma_{t-m}^2 + e_t$$

where e_t denotes an error term. Thus the conditional variance of r_t would be given as:

$$E(\sigma_t^2 | \sigma_{t-1}^2) = \alpha_0 + \alpha_1 \sigma_{t-1}^2 + \cdots + \alpha_m \sigma_{t-m}^2$$

It is also possible to write the process of a_t in a multiplicative form; an ARCH(1) model is given below:

$$r_t = \varphi_0 + \varphi_1 r_{t-1} + a_t$$
$$a_t = \sigma_t \varepsilon_t$$
$$\sigma_t^2 = \alpha_0 + \alpha_1 \sigma_{t-1}^2$$

where ε_t is a variable from a white noise process, $\alpha_0 > 0$, and $\alpha_1 >= 0$.

The unconditional mean of a_t remains 0:

$$E(a_t|\sigma_{t-1}) = E[\sigma_t E(\varepsilon_t)] = 0$$

Because a_t is a stationary process, the unconditional variance of a_t can be shown as:

$$Var(a_t) = E(\alpha_0 + \alpha_1 \sigma_{t-1}^2) = \alpha_0 + \alpha_1 Var(a_t)$$
$$\Rightarrow Var(a_t) = \frac{\alpha_0}{1 - \alpha_1}$$

The ARCH model has the following properties:

- Historical data often show that a recent period of high volatility will normally be followed by periods of high volatility, and vice versa. This is known as the volatility clustering effect. Note that if σ_{t-1}^2 is large, σ_t^2 is also large. Hence the model above is consistent with this observation.
- The distribution of a_t shows excessive kurtosis (fat tail), which means that the ARCH model can more likely produce "outliers", i.e. abnormal movements, which is often seen

in financial time series. However, financial time series often respond differently to positive and negative shocks, whereas in the ARCH model these shocks are treated in the same way (since it is a square model)

- The model tends to react slowly to changes (i.e. it has a long "memory"), which may be a desirable or undesirable property.

GARCH model

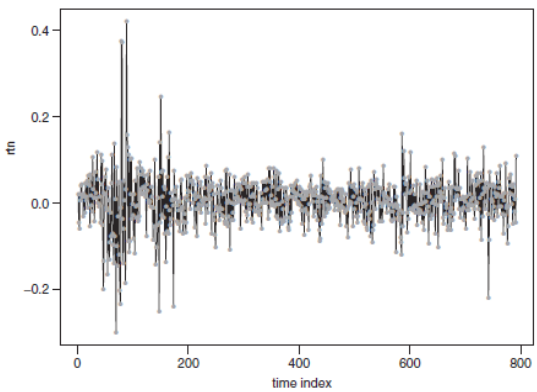
Instead of using an AR process for σ_t^2, Bollerslev (1986) proposes to use an ARMA model:

$$a_t = \sigma_t \varepsilon_t$$
$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{m} \alpha_i a_{t-i}^2 + \sum_{j=1}^{s} \beta_j \sigma_{t-j}^2$$

In the case of m=s=1, we have the GARCH(1,1) specification. $\sigma_t^2 = \alpha_0 + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2$
Alternatively, we can write it in the form

$$\sigma_t^2 = \gamma V_L + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2, \gamma + \alpha + \beta = 1$$

where V_L is the long run average variance rate.



$$r_t = 0.0076 + a_t$$
$$\sigma_t^2 = 0.000086 + 0.1216 a_{t-1}^2 + 0.8511 \sigma_{t-1}^2$$

- Equivalent long run annualized volatility = 19.47%

As an example, the graph above shows the monthly excess returns of S&P 500 index from 1926 to 1992 (with 792 observations). Note that the final model is without the AR parameters for the returns r_t. (Example taken from Tsay, 2nd edition, p.117-118)

GARCH(1,1) can be used in volatility forecasting. With

$$\sigma_t^2 = \gamma V_L + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2 = (1 - \alpha - \beta) V_L + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2$$

On day t+n in the future,

$$\sigma_{t+n}^2 - V_L = \alpha(a_{t+n-1}^2 - V_L) + \beta(\sigma_{t+n-1}^2 - V_L)$$

Since expected value $E[a_{t+n-1}^2] = \sigma_{t+n-1}^2$, thus

$$E[\sigma_{t+n}^2 - V_L] = (\alpha + \beta) E[\sigma_{t+n-1}^2 - V_L] = (\alpha + \beta)^n (\sigma_t^2 - V_L)$$
$$E[\sigma_{t+n}^2] = V_L + (\alpha + \beta)^n (\sigma_t^2 - V_L)$$

Using a numerical example, if $\alpha+\beta = 0.9602$ and $V_L = 0.00004422$, the expected variance in n=10 (days) $= 0.00005473$. The expected variance in n=100 (days) $= 0.00004449$, which is very close to V_L, the long term expected volatility.

GARCH can be used in the context of derivative modelling and risk management. Assume that we adopt the market implied volatilities in our pricing model. However, when volatility moves, it may not move evenly across the whole term structure. In particular, it is often observed that long term volatilities may move less than short term volatilities. We may use the GARCH predictions to work out a "weighted vega". The following example is taken from Hull (5th edition, 2003), p.385. The initial instantaneous volatility is 12.30%, and it is changed to 13.30%. Instead of applying a customized change, note how the model can be used to generate a plausible scenario:

Impact of 1% change in the instantaneous volatility predicted from GARCH (1,1)					
Option life (days)	10	30	50	100	500
Option volatility now (%)	12.03	11.61	11.35	11.01	10.65
Option volatility after change (%)	12.89	12.25	11.83	11.29	10.71
Increase in volatility (%)	0.86	0.64	0.48	0.28	0.06

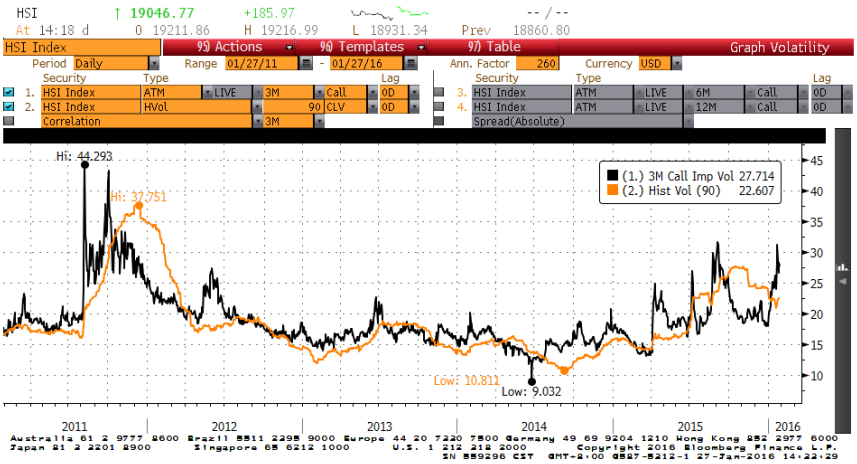
8.2.2. Trading volatility as an asset class

Sophisticated investors such as hedge funds may want to take a trading view on the volatility of some underlying variables. Let's say the current historical volatility of Hang Seng Index is 30%, and a trader thinks that it would increase to 35% before the end of 2024. How can the trader design a trading strategy to make money if this view is right?

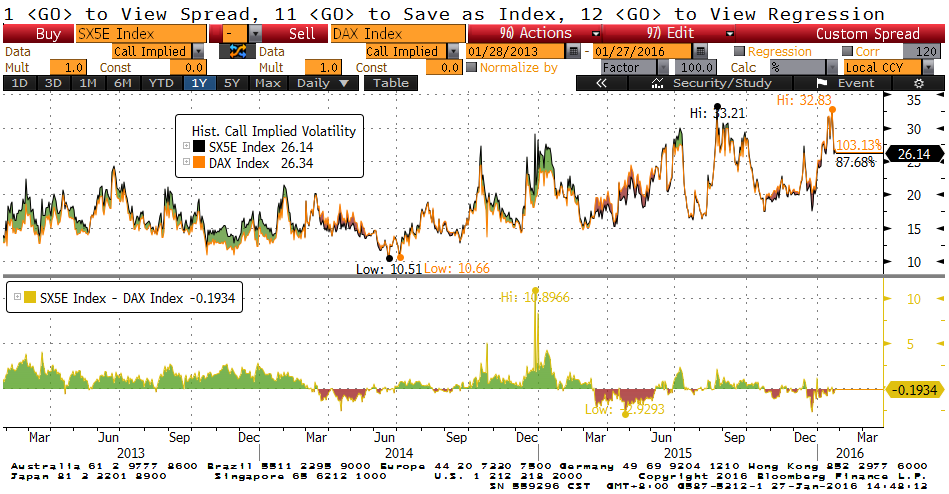
If we can only trade instruments based on the index, some kind of model is needed to relate the movement of the index to its volatility. Alternatively, markets have been developed, so that some forms of the volatility become underlyings themselves which can (almost) be traded directly.

There are three main kinds of volatility trading strategies:

1. A pure play on the implied volatility of one single underlying.
2. Trades to capture the perceived difference in the implied volatility and the realized volatility of a single underlying. The graph below shows the spread of the realized and implied volatilities of the Hang Seng Index between 2011 and 2016 (source: Bloomberg). It is observed that during normal periods, implied volatility is often (slightly) higher than the realized historical volatility.



3. A spread trade to capture the difference in the implied volatilities of two different underlyings. The graph below shows the spread of the implied volatilities between Eurostoxx 50 and DAX options between Jan 2013 and Jan 2016 (source: Bloomberg). Given the variation, is there any trading opportunity?



8.2.2.1 Trading through options

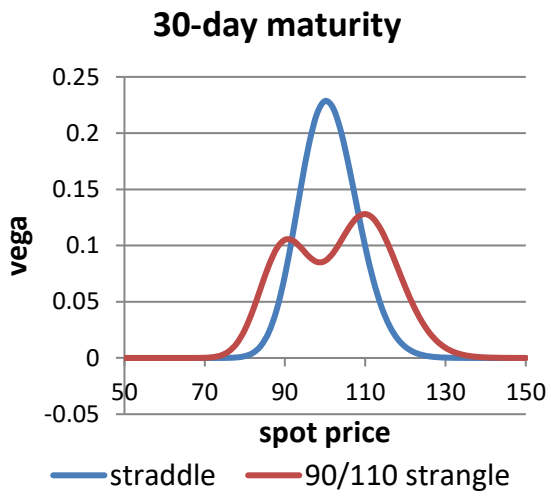
Trading vanilla options is often the first choice in trading volatility. Firstly, options have vega exposures, and can be used to take positions in the implied volatility. Secondly, through delta hedging, the difference between realized volatility and implied volatility can be captured.

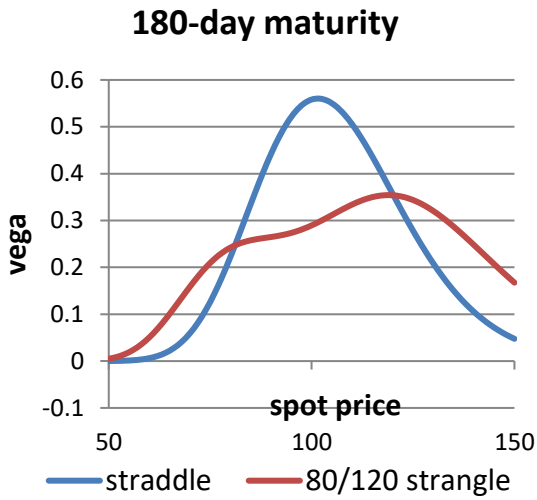
If we want to use vanilla options to take advantage of the change in implied volatilities, it is important to hedge the other risks of the positions. We start with a simple example:

	Spot price	Volatility	Option price
(1)	100	30%	13.282
(2)	100	35%	15.216
(3)	95	35%	12.345

This table shows the changes in the theoretical prices of a European call option, 1 year maturity, rate 3%, strike 100. A long position in a vanilla option has a positive vega. When volatility goes up, option price should go up. If we are long the 100 strike call option and volatility goes up when spot price stays the same (case (2) above), we make money. But from case (1) to case (3), volatility goes up from 30% to 35%, while spot price drops from 100 to 95, and a long position in this option will still lose money! The loss comes from the delta risk, but other risks, such as interest rate exposure, should also be hedged in order to capture the pure effect of volatility changes.

Another important characteristic of vanilla options is the change in vega when spot price changes. In section 8.1.3.4, we have seen that the vega of a position is highest when the spot price is near to the at-the-money level. Section 8.1.4.3 gives an example of rapid vega changes when the spot price moves away from the strike. One way of retaining enough vega exposure is to use options with longer maturities. Alternatively, a strategy such as a straddle or a strangle is useful.





The above diagrams show the vega exposures of a straddle (long call and put, both at strike K=100) and a strangle (long call strike K_1, long put strike K_2, $K_1 > K_2$). In these examples, we note that a strangle can retain certain vega over a wide range of strikes, which can be useful for trading volatility.

Single option or strangle/straddle can be used to capture the anticipated direction movement of volatility of the underlying. But it is also possible to design a trading strategy to reflect the change in the shape of the volatility skew (c.f. section 8.2.3). This strategy mostly involves the trading of options on equity indices, and may even involve high frequency trading of listed options. For example, the difference between the volatility of an at-the-money option and a 110% strike option is 2%, and the trader's view is that this difference will increase to 3% within one week. A spread trade in options in the various strikes, together with the proper delta hedge, should be entered.

8.2.2.2. Listed instruments with (implied) volatility as the underlying

As in the case of other futures contracts, the contracts should be based on certain kinds of underlying reference price levels. The most popular listed instruments would make use of volatility indices, which are derived from the implied volatilities of listed options. Two

examples are the VIX futures and the HVI futures, based on the S&P 500 and the Hang Seng Index.

VIX index

Source: CBOE

Introduced in 2004, VIX is an aggregate index representing a weighted average of the volatilities of the skew of listed S&P 500 options with 30-day maturity at different strikes.

$$\sigma_{VIX}^2 = \frac{2}{T}\sum_{i=1}^{N}\frac{\Delta X_i}{X_i^2}exp(rT)V(X_i) - \frac{1}{T}\left(\frac{F}{X_0} - 1\right)^2$$

$$F = X_0 + exp(rT)(C_0 - P_0)$$

$$\Delta X_i = \frac{X_{i+1} + X_{i-1}}{2}$$

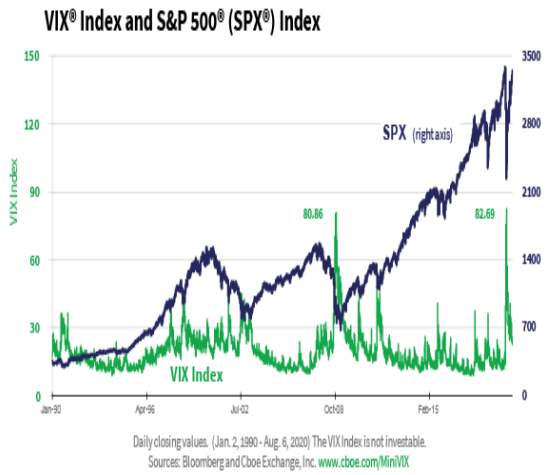
The above formula is applied to calculate an aggregate volatility for options expiring at a particular time T. r is the risk-free rate, T is the expiration time, F is the forward price of the index, X_0 is the strike immediately below the forward price, X_i is the strike of the ith out-of-the-money option, V is the mid-price of the corresponding option. Note that F is derived from put-call parity, through the available market prices of call C_0 and put P_0.
The definition of VIX requires a fixed maturity of 30 days. Hence we need to apply a certain interpolation rule to arrive at this aggregate. First, we apply the above formulas to the first two option expirations, T_1 and T_2, where T_1 <= 30 days <= T_2. We can then interpolate to find a constant 30-day volatility:

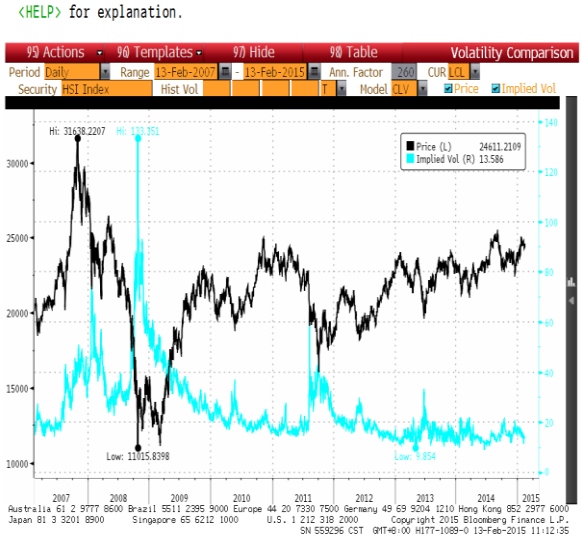
$$VIX = 100\sqrt{\left[T_1\sigma_{VIX1}^2\left(\frac{N_{T_2} - N_{30}}{N_{T_2} - N_{T_1}}\right) + T_2\sigma_{VIX2}^2\left(\frac{N_{30} - N_{T_1}}{N_{T_2} - N_{T_1}}\right)\right]\frac{N_{365}}{N_{30}}}$$

Note that N_T is the number of minutes remaining for the appropriate contract.

VIX is often known as the “fear index”, as its level will often increase when there is a severe downturn in the underlying market. Historically, we could often observe a negative correlation between the underlying price and the ATM volatility of options, especially when

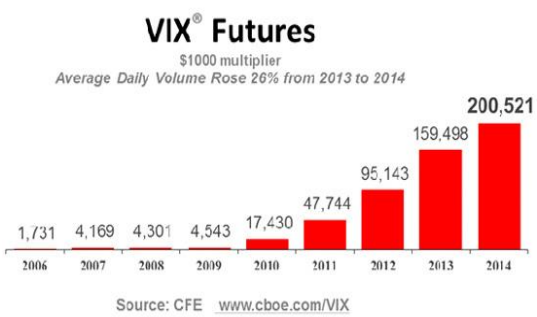
the index is range bound. The left diagram below is the levels for SPX from Jan 1990 to Aug 2020, and the right diagram is HSI from Feb 2007 to Feb 2015.

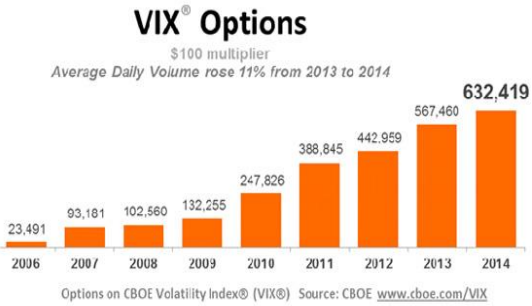




VIX futures and options

The VIX is not a tradable asset by itself, as it is a sliding measure of implied volatility. Historically, the VIX was developed to facilitate trading of listed instruments, including futures and options, based on this index. Basically, futures on VIX is a pure play on (forward) volatility level independent of the direction and level of the stock price index (S&P 500). Trading of VIX-based products have been very active in recent years. (Further data can be obtained from the VIX fact sheet from CBOE, https://cdn.cboe.com/resources/vix_options/VIX_fact_sheet.pdf).



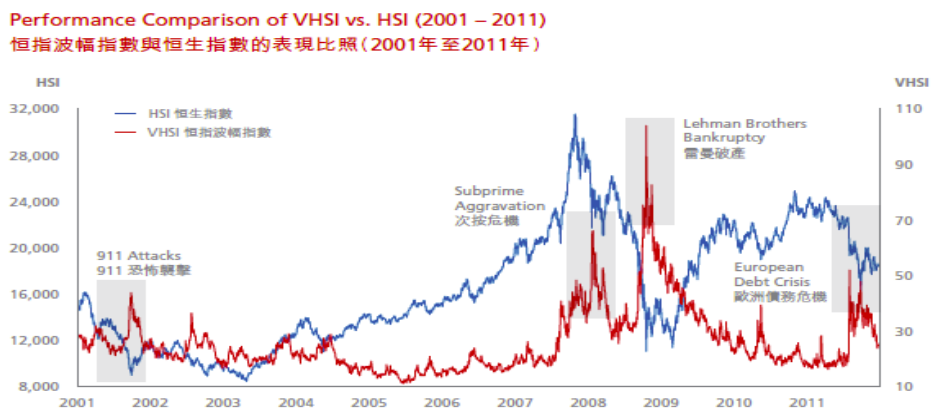


In a VIX futures contract, each point represents US$1000, and the minimum movement is a 0.05-point step. Maturities can be up to 9 near-term serial months and five months on the February quarterly cycle. For example, if the April 2022 VIX futures is trading at 31.80, it means that the market expectation for the average volatility of options with a 30-day maturity as seen in April 2022 is 31.80%. The table below is a snapshot of VIX futures on March 4, 2022 (source: CBOE). Note how the term structure varies.

VIX

SYMBOL	EXPIRATION	LAST	CHANGE	HIGH	LOW	SETTLEMENT	VOLUME
VIX	-	32.01	1.50	34.65	34.65	-	-
VX10/H2	03/09/2022	0		0	0	31.90	0
VX/H2	03/15/2022	33.45	1.923	34.15	32.40	31.527	6,904
VX12/H2	03/23/2022	0		0	0	29.75	0
VX13/H2	03/30/2022	0		0	0	30.00	0
VX14/J2	04/06/2022	0		0	0	29.90	0
VX15/J2	04/13/2022	0		0	0	29.90	0
VX/J2	04/20/2022	31.80	1.499	32.05	31.00	30.301	2,133
VX/K2	05/18/2022	30.45	1.139	30.65	29.88	29.311	1,023
VX/M2	06/15/2022	29.50	0.864	29.68	29.02	28.636	841
VX/N2	07/20/2022	29.07	0.577	29.32	28.77	28.494	265
VX/Q2	08/17/2022	28.45	0.551	28.60	28.15	27.899	192
VX/U2	09/21/2022	28.10	0.369	28.30	28.05	27.731	65
VX/V2	10/19/2022	27.85	0.275	28.00	27.85	27.575	4
VX/X2	11/16/2022	27.18	0.355	27.18	27.18	26.825	1

A contract similar to the VIX futures was introduced in Hong Kong in February 2012. The contracts are based on the VHSI index, which is a volatility benchmark for the HSI, similar to the calculation methodology of VIX. Each point movement of the futures price represents HK$5000. However, up to now, trading has been inactive and there had been no interest in the product currently.



VIX trading strategies

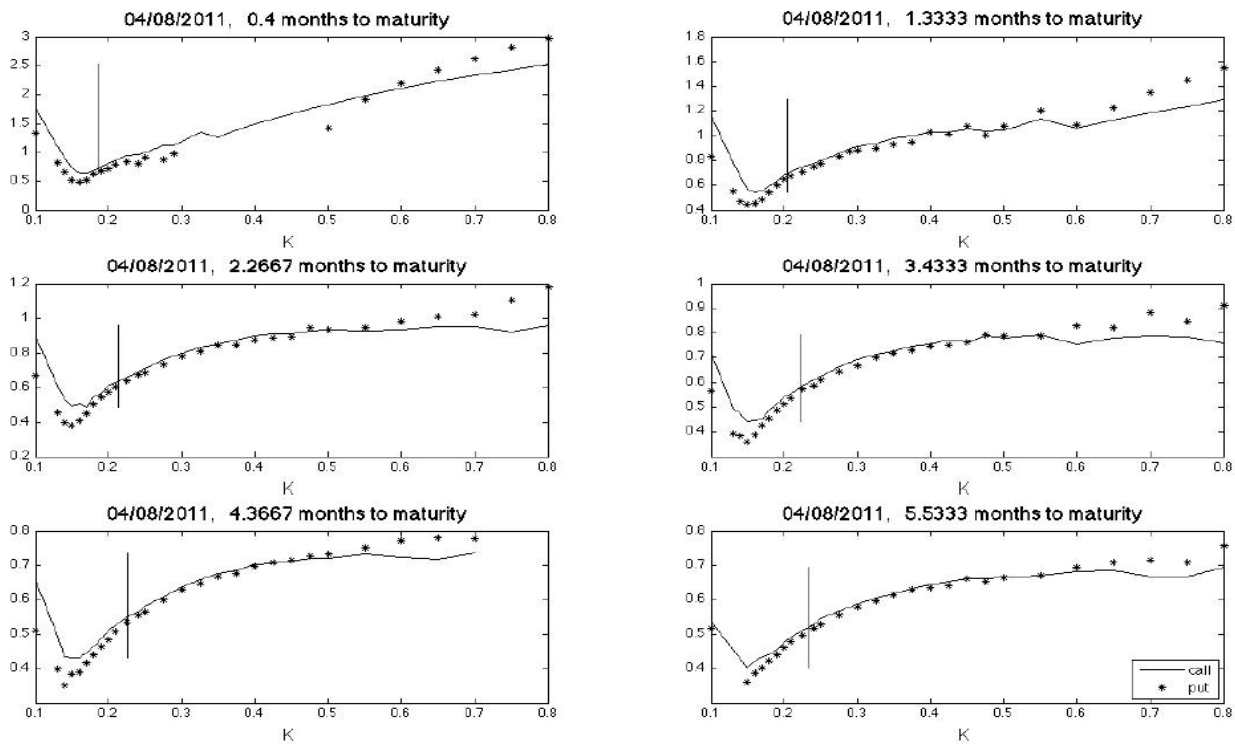
A general feature of the VIX term structure is that it tends to mean revert: it is downward sloping when VIX is high (say > 40), whereas it is upward sloping when VIX is below 20. Given this observation, it is still difficult to have a position in one maturity that can be hedged with a position in another maturity.

In a study by Simon and Campasano (2013) using data from 2006 and 2011, they examine the predictive power of the VIX futures basis, defined as the difference between the VIX and the price of the first futures contract. They found that the basis does not predict changes in the level of VIX. However, it was found that a successful trading strategy is:

1. Define the daily roll as the difference between the front month VIX futures and the cash VIX, divided by the number of business days to settlement.
2. Short the future if the price is above the cash, and the daily roll is greater than 0.1 point.
3. Long the future if the price is below the cash, and the daily roll is less than –0.1 point.
4. Hold the trade for five days.
5. Since there is often a strong negative correlation between the changes in volatility and the returns of S&P 500, a partial hedge to the spot risk is achieved via having a long position in S&P 500 futures if the trade is long VIX futures, and short S&P 500 futures if the trade is short VIX futures, with a hedge ratio obtained through regression.

Analysis and trading of VIX futures can be carried out in two different ways. As we note above, VIX is not a single underlying that can be traded. Theoretical prices of VIX futures can only be obtained by assuming a model for the future evolution of implied volatility of the S&P 500 index itself. Without such a model, there is no consistent way to relate the contracts at different maturities.

A simplified approach may treat the term structure of VIX futures as inputs for calibration of a model based on VIX as the underlying itself. If we have a model on VIX, options on VIX can be priced, and the volatility skew on this underlying can be analyzed.



The diagrams above (from Papanicolaou and Sircar (2013)) come from the market prices of vanilla calls and puts on the VIX. They are commonly analyzed through the Black's model in order to compute "implied volatilities" of the VIX.

Of course, the ultimate modelling goal is to be able to specify the dynamics of the S&P 500 index itself that can be consistent with the market prices of the index options, as well as capturing the prices of the VIX futures and options. Such a model will be able to manage the risks of positions relating to any kind of derivative of the S&P index. The current state of the VIX and SPX markets do not seem to converge, and much work need to be done in aligning the two markets together.

8.2.2.3. Through an OTC instrument with volatility as the underlying

Given the high demand of products based on volatility, many over-the-counter (OTC) instruments were developed to satisfy the customer needs. The first significant product, known as variance swap, was first traded in the late 1990s. Other products, such as volatility swaps, options on variance, and gamma swaps, have appeared in the market, but most of these are traded with the biggest names / indices only. Some of these products are based on the difference between realized and implied volatilities of the underlying assets, whereas others are pure play on the implied volatility.

Variance swaps

A variance swap, as its name suggests, is a bet on variance (i.e. square of volatility). It is actually a forward, not a swap, as there is no exchange of cashflow during the life of the swap. The final payoff depends on realized variance of the reference index or asset. There has been active trading for many stock market indices. For example, S&P 500 variance swaps can trade the equivalent of $5 million per vega point, with a bid-ask spread of 0.30% in volatility quote. The fair price takes into account of the whole skew, similar to the definition of the VIX contract discussed in the section above.

A typical variance swap has the following terms:

Maturity:	1/14/2015
Underlying:	Nikkei 225 index
Variance amount:	473,933.65 (= A)
Vega notional:	JPY 20,000,000
Volatility strike:	21.1
Variance strike:	445.21 (= 21.1^2)
Payoff:	A x (realized variance – variance strike)

- The actual payoff is based on the variance, although it is common for a volatility strike to be quoted.
- *Variance amount* is the multiplier used in calculating the actual payoff.
- *Vega notional* is not used in the payoff; it is roughly the P&L when the realized volatility moves by 1%.
- The relationship of the various terms are illustrated below:

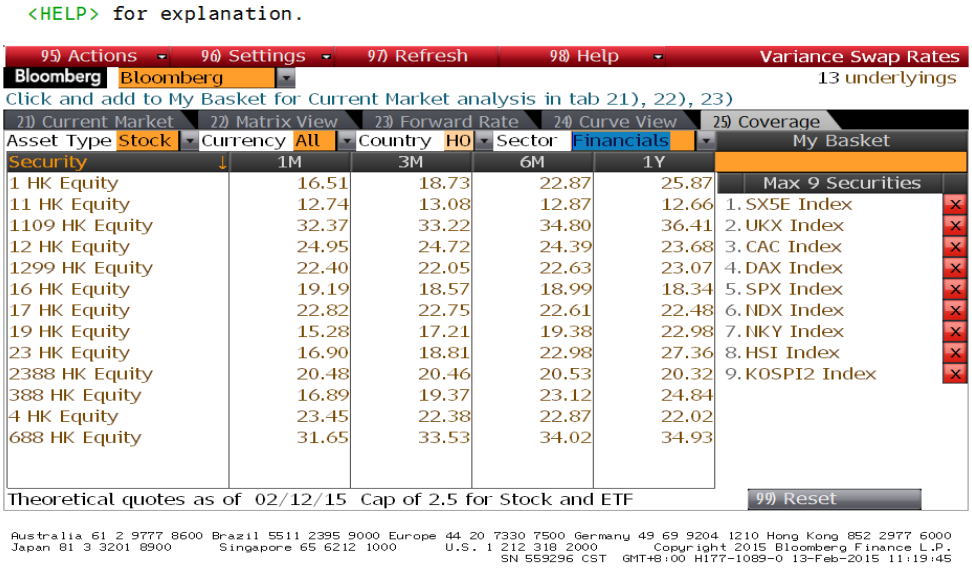
  Variance amount = vega notional / 2 / volatility strike
  E.g. variance amount = 20000000 / 2 / 21.1 = 473933.65
  For a 1% move in volatility (from 21.1% to 22.1%), P&L is:
  473933.65 x ($22.1^2 – 21.1^2$) = JPY 20,473,933
  For a 1% move in volatility (from 21.1% to 20.1%), P&L is:
  473933.65 x ($20.1^2 – 21.1^2$) = JPY –19,526,066
- It is evident that variance swap is a non-linear product with respect to volatility.
- Realized variance is usually calculated by the following formula,

$$\sigma^2 = \frac{252}{n-1}\sum_{i=1}^{n-1}\left(ln\frac{P_{i+1}}{P_i}\right)^2 \times 10000$$

n is the number of business days between trade date and maturity, P_i is the daily closing price of N225 on the ith business day, P_n is the closing price of the index on the maturity date.

This formula is slightly different from the typical formula for calculating volatility, where we need to account for the mean. The reason is because while omitting the mean will have little impact on the final value, it will make the calculation of payoff perfectly additive, so that the calculation of payoff in early closeout will be much simpler, as shown in the following example.

Some market quotes of variance swap strikes of different Hong Kong listed stocks are shown below (source: Bloomberg, as of 2/12/2015). A term structure is clearly seen for some of the underlyings, reflecting the term structure of volatilities, as discussed in the following sections on pricing.



Variance swap payoff calculation examples

Let's say we enter into a 1-year variance swap on day 1, receiving the variance and paying a variance strike of 400 on S&P500. After 3 months, we enter into a 9-month variance swap, paying the variance and receiving a variance strike of 576. The purpose of this trade is to lock into the profit of the previous trade.

If the variance payment is defined in the standard way, i.e. variance = $\Sigma(x-\mu)^2$, we can see that the payments in the first swap cannot net off the payments in the second swap, as μ is different in the two cases (one is based on the average of 1 year variance, the other is the average of 9 months variance). However, if the variance payment is defined as Σx^2, the payments in the second swap would become identical to that of the first swap, leaving the net floating payments to be coming from the first 3 months only.

variance units		6704.32
variance strike	=35^2	1225
position		short
no. of fixings		7
scaling		252

At maturity

		close	ln(return)	ln(return)^2
	10/12/2007	17,187.28		
1	11/12/2007	17,497.77	0.017904	0.000321
2	12/12/2007	16,962.62	-0.031061	0.000965
3	13/12/2007	16,332.08	-0.037881	0.001435
4	14/12/2007	15,957.46	-0.023205	0.000538
5	17/12/2007	15,391.39	-0.036118	0.001305
6	18/12/2007	15,449.47	0.003766	0.000014
7	19/12/2007	15,638.68	0.012173	0.000148
sum				0.004726
scaled		=sum*252/7		0.170124
calculated variance				1701.237095
payoff		= 6704.32 x (1225 - 1701.237095)		

variance units		6704.32
variance strike	=35^2	1225
position		short
no. of fixings		7
scaling		252

On 12/14/07

		close	ln(return)	ln(return)^2
	10/12/2007	17,187.28		
1	11/12/2007	17,497.77	0.017904	0.000321
2	12/12/2007	16,962.62	-0.031061	0.000965
3	13/12/2007	16,332.08	-0.037881	0.001435
4	14/12/2007			
5	17/12/2007		contributions from	
6	18/12/2007		implied vol between	
7	19/12/2007		12/14/07 - 12/19/07	
sum				0.002720
scaled		=sum*252/7		0.097931
calculated variance				979.312596
			+contributions from	
			implied vol	

The left table shows the calculation of the final payoff of a typical variance swap. The trade is a short position in a 7-day variance swap, with variance strike = 35^2 = 1225, and variance amount = 6704.32. The right table shows how the intermediate mark-to-market value can be calculated, which is useful if there is a need for an early close-out of the trade. The fixings from 11/12/2007 to 13/12/2007 are known. As of 14/12/2007, the contributions for the remaining portion of the variance swap need to derive as if a new 4-day variance swap is started on that day. It means that the fair level would come from the implied volatility skew of the underlying (as shown in the pricing section below). The total mark-to-market value of the swap as of 14/12/2007 is thus the simple sum of the already fixed portion together with the portion coming from the implied volatility skew.

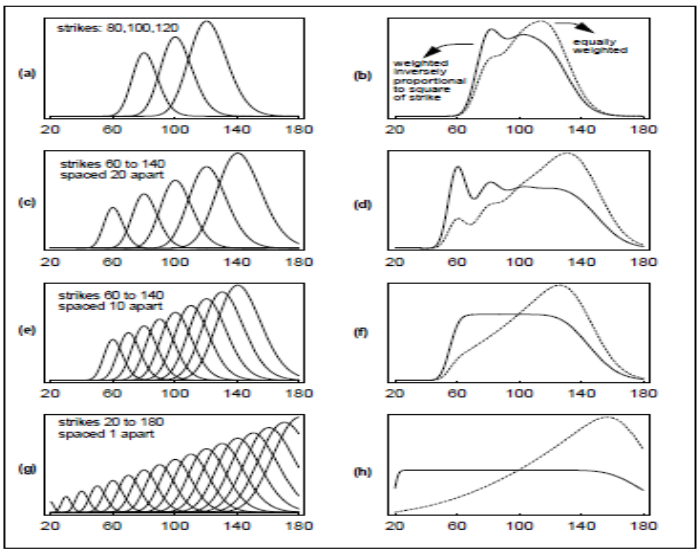
Variance swap: theoretical considerations

In section 8.2.2.1, we have considered how vanilla options can be used to engage in volatility trading. However, one drawback of the approach is that the sensitivity of volatility of the position is often affected by the moneyness. When the spot price moves away from the strike, the position loses its sensitivity to further movements in volatility. Long dated strangles may work better, but would still require an active re-hedging of the other risks (such as delta risks) in order to benefit from the volatility movements.

On the other hand, variance swap allows investors to take a pure volatility view that is not affected by the underlying level relative to a particular strike/strikes. It would always retain an exposure to volatility even though underlying levels may have moved dramatically. It has the further advantage that the strike of the swap relates to the shape of the volatility skew for options with the same maturity, and not only to an option with a particular strike.

If we assume that jumps do not occur and interest rates are deterministic, mathematically there is an exact replicating strategy for variance swap, whereas there is no

explicit strategy for volatility swap, therefore variance swap used to be a far more popular product (but see the discussion in the section below).



What kind of replicating strategy will be appropriate? From Demeterfi et al. (1999), the first diagram on the left shows the variance sensitivity profiles of three options, with strikes 80, 100, and 120, plotted on the same scale. If we weigh them equally and add them together to form a single portfolio, the resulting variance sensitivity profile is plotted in the right diagram. It also shows the result of a portfolio that weighs the options inversely proportional to the square of the strike levels, i.e. weights of $1/K^2$. The diagrams below show the cases when more options are added. From these diagrams, we can see that if options of all strikes are available and weighted by $1/K^2$, the resulting variance sensitivity profile is almost flat. This is the intuitive replication portfolio of the variance swap. However, note that since more OTM puts are needed than the calls, the position is long volatility skew (c.f. section 8.2.3).

Brief note on variance swap pricing

When variance swap first became popular in the late 1990s, the calculation of the fair strike was claimed as a "model independent valuation" as an elegant derivation shows that the pricing only requires the availability of vanilla option prices at various strikes. Following Demeterfi et al. (1999), if the underlying follows a diffusion process (note that μ and σ can be functions of S and t),

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW$$
$$d(log\, S_t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW$$
$$\Rightarrow \frac{dS_t}{S_t} - d(log\, S_t) = \frac{1}{2}\sigma^2 dt$$

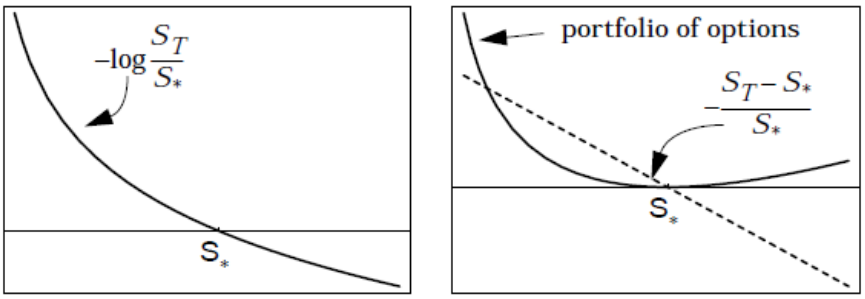
The fair strike of the variance swap is given as the expected value of the realized variance:

$$K_{var} = E[V] = \frac{1}{T}E\left[\int_0^T \sigma^2 dt\right] = \frac{2}{T}E\left[\int_0^T \frac{dS_t}{S_t} - log\frac{S_T}{S_0}\right] = \frac{2}{T}\left[rT - E\left[log\frac{S_T}{S_0}\right]\right]$$

The key is to replicate the log contract. For an arbitrary S_* (such as the forward price at time T), the log contract can be written as:

$$log\frac{S_T}{S_0} = log\frac{S_T}{S_*} + log\frac{S_*}{S_0} - log\frac{S_T}{S_*}$$

$$= -\frac{S_T - S_*}{S_*} + \int_0^{S_*} \frac{1}{K^2}\text{Max}(K - S_T, 0)dK + \int_{S_*}^{\infty} \frac{1}{K^2}\text{Max}(S_T - K, 0)dK$$



The fair strike of the variance swap is thus:

$$K_{var} = \frac{2}{T}\left[rT - \left(\frac{S_0}{S_*}e^{rT} - 1\right) - log\frac{S_*}{S_0} + e^{rT}\left(\int_0^{S_*} \frac{1}{K^2}P(K)dK + \int_{S_*}^{\infty} \frac{1}{K^2}C(K)dK\right)\right]$$

where $P(K)$ and $C(K)$ are the current fair values of a put and call option at strike K. For initial pricing, the expression can be simplified if we choose S_* as the forward of S_0, given by $S_* = S_0e^{rT}$. Note that this derivation only requires the underlying to follow a diffusion process (jumps will be discussed below).

Approximate formulae for variance swap strike

Instead of utilizing all the options of the same maturity, Demeterfi et al. (1999) give a rule of thumb for the fair strike, applicable when the skew is linear in strike:

$$K_{var} = \sigma_{ATM}\sqrt{1 + 3T \times skew^2}$$

where σ_{ATM} is the at-the-money volatility, T is the maturity, and *skew* is the slope of the skew curve.

For log-linear skew (which is more realistic), the rule is:

$$K_{var} = \sqrt{\sigma_{ATM}^2 + \beta\sigma_{ATM}^3 T + \frac{\beta^2}{4}(12\sigma_{ATM}^2 T + 5\sigma_{ATM}^4 T^2)}$$

The skew is assumed to be: $\sigma(K) = \sigma_{ATM} - \beta\, ln(K/forward)$.
Note that these two rules only produce good results when the skew is not steep.

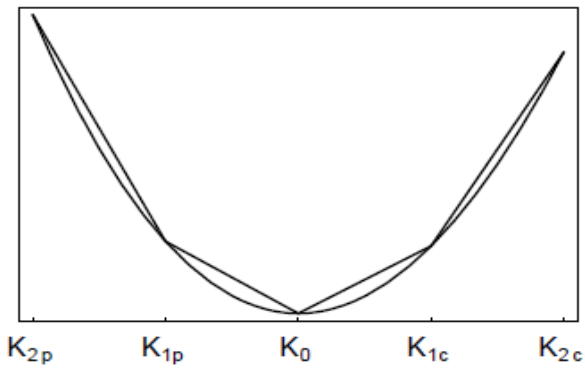
Hedging the variance swap

K_{var} can be replicated by the following:

- Short $2/T$ x $\log(S_*/S_0)$] units of bond.
- Short $2/TS_*$ units of forward.
- Long $2/(TK^2)$ units of puts for strike below S_*, and calls for strike above S_*.
- Dynamically re-balance the portfolio to hold $2/(TS)$ shares everyday, so that the value of this position is a constant at $\$2/T$.

Given that it is not feasible to hedge with a continuous range of strikes, an approximation is to hedge with a portfolio Π_{CP} that exceed the payoff of the log contract, such that

$$\Pi_{CP} = \sum_i w(K_{ip})P(S, K_{ip}) + \sum_i w(K_{ic})C(S, K_{ic})$$



In order to hedge the curved profile, the proportions of calls that would be held can be:

$$w_c(K_0) = \frac{f(K_{1c}) - f(K_0)}{K_{1c} - K_0}$$

$$w_c(K_{n,c}) = \frac{f(K_{n+1,c}) - f(K_{n,c})}{K_{n+1,c} - K_{n,c}} - \sum_{i=0}^{n-1} w_c(K_{i,c}),\ n \geq 1$$

A similar expression can be derived for the puts. Note that the hedging error is smaller when the strikes are closer together.

In the presence of jumps, the replication strategy can exhibit higher hedging error. It can be shown that the P&L of the replication strategy using the strip of calls and puts is of the order of:

$$\text{P\&L due to jump} = \frac{2}{3}\frac{J^3}{T} + \cdots$$

where J is defined as the % jump downwards, from S to $S(1-J)$ (Demeterfi et al. (1999); c.f. Gatheral (2006), chap. 11). Therefore a large jump move downwards ($J>0$) leads to a profit for the short variance swap position (hedged with the replication strategy given above). Furthermore, a (small) jump one day followed by a (small) jump in the opposition direction

the next day would tend to offset each other. When jumps are large, the 4th order terms become significant and cannot be ignored. Also, jumps may cause the spot price to move outside of the range of strikes used in the replication strategy described above.

Variance swap greeks

The typical greeks of the variance swap are very different from those of vanilla options. Their behaviour are also different, and other sensitivity parameters provide additional information. We already note that a simple delta hedge position is adopted, so that the dollar value of the share position is always equal to $\$2/T$. Other standard Greek parameters are: $\text{gamma} = \frac{2}{T}\frac{1}{S^2}$, $\text{vega} = \frac{2}{T}\sigma(T-t)$. The following points are noted:

- Cash gamma, which is defined as the gamma multiplied by S^2, is constant. This term is independent of time to maturity and underlying price.
- Vega is linear with σ, but shows a positive vega convexity
- Variance vega, which is the sensitivity to a move in the variance, is decreasing according to the time to maturity but is not sensitive to the underlying price. It is given as:
$$\text{variance vega} = \frac{T-t}{t}$$
- $\text{theta} = -\frac{1}{T}\sigma^2$. It is a constant, i.e. the variance swap loses value at a constant rate if stock price doesn't move.

There are two sensitivities which are related to the volatility skew:

- Skew delta – when stock price goes down, implied volatility will go up because the ATM volatility will go down (c.f. next section).
- Vega convexity – this is the change in vega when the volatility changes; hence when stock price goes down and implied volatility goes up, the vega will also go up. It is given as:
$$\text{vega convexity} = 2\frac{T-t}{t}$$

We show a step-by-step calculation of the sensitivities, using the earlier example (in the P&L calculation section): short position in a 7-day variance swap, with variance strike = 35^2 = 1225, and variance amount = 6704.32.

Example 1: (left table) sensitivity to underlying movement

- From the calculation of the realized variance, it is evident that the contribution of each day has the same weighting, depending on the number of fixing days. Therefore, if there are 7 fixing days, each day contributes to 1/7 of the total calculated variance in this example.
- Hence, the delta is the same everyday. For example, if the move in one of the days is 2%, the contribution to the variance is: $[\text{Ln}(1.02)]^2 * 252 / 7 * 10000 = 141.1719$. If the move is 4%, the contribution is: $[\text{Ln}(1.04)]^2 * 252 / 7 * 10000 = 553.7752$
- But it is noted that the move is non-linear, i.e. it is long gamma (if we are long variance swap).
- If there are more fixing days, each day's move is less significant.

Example 2: (right table) sensitivity to volatility movement

- Let's say the implied volatility is equivalent to a move of 2% per day (approx. 32% volatility), so the four days between 12/14/07 and 12/19/07 would each be represented by a variance contribution of 141.17 (as calculated above).
- At the close of 12/13/07, the MTM fair price is the sum of the contribution from historical moves (979.31) + contribution from the implied volatility (141.17 x 4) = 1543.99.
- Alternatively, if the implied volatility is equivalent to a move of 2.5% per day (approx. 40% volatility), the daily contribution is: $[\text{Ln}(1.025)]^2 * 252 / 7 * 10000 = 219.50$.
- At the close of 12/13/07, the MTM fair price is 979.31 + 219.50 x 4 = 1637.81.
- Clearly, vega depends on the number of days between now and maturity (=4 in the above example) – vega would become smaller and smaller as the contract approaches expiry. Reduction of vega is linear with respect to time to maturity (c.f. vanilla options) and it is not affected by the current spot level.

Variance swap before and after 2008

The financial instrument that investors really want is one that is based on volatility. However, before 2007, variance swaps based on both equity indices and single stocks were more popular because it was assumed that a perfect hedge is theoretically possible. In reality, investment banks had short position whereas hedge funds were buyers, and since the deep out-of-the-money options may not be liquid, the investment banks were short the tail risk. During the financial crisis of 2008, the unhedged portions were often reached, leading to substantial losses by the investment banks. As a result, after 2008, volatility swap becomes popular again because it is seen that the "perfect hedge" argument is an illusion anyway.

For single stock variance swaps, there is an additional level of complexity as it is always possible that the company could go into default, and hence the variance can become undefined (when stock price drops to 0). Therefore, a single stock variance swap is often capped, typically at 250% to 400% of the variance strike, i.e.

$$P = \min(Cap, \sigma_R^2) - K_{var} = \sigma_R^2 - K_{var} - \max\left(\sigma_R^2 - Cap, 0\right)$$

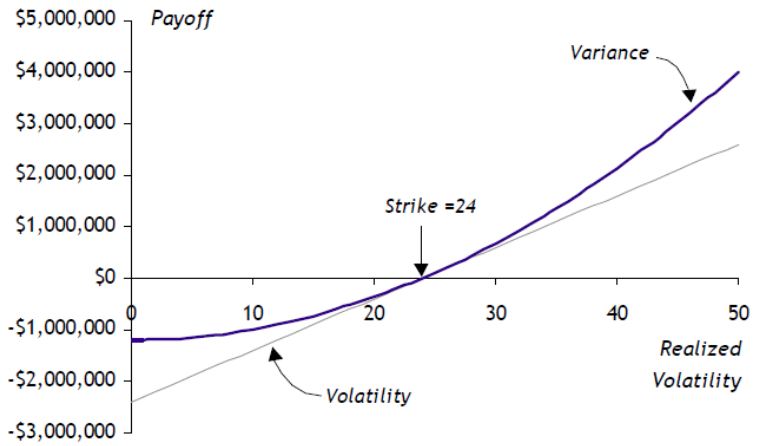
Theoretically, the pricing of a variance swap with cap requires the pricing of an option on realized variance. Many of these caps were left unhedged, but they were hit during the financial crisis in 2008.

Variance swaps and volatility swaps

When variance swap and volatility swap of the underlying are available, the strike levels are different. As in the example above, variance swap is quoted as the square root of the variance in order to allow easier comparison with implied volatility. However, the variance swap quote is higher than the corresponding vol swap quote, whereas the volatility swap quote is close to the at-the-money volatility of the underlying. This is evident from Jensen's inequality:

$$K_{vol} = E\left(\sqrt{\sigma_{realized}^2}\right) \leq \sqrt{E(\sigma_{realized}^2)} = K_{var}$$

Another explanation is due to the convex payoff of the variance swap. Only in the case when the volatility of volatility is 0 (i.e. future volatility is constant and known) will the price of the two be the same. A typical variance swap payoff is shown below (from Bossu et al. (2005)):



Gamma swaps

Instead of applying a constant weighting, a weight function is applied to the payoff of the variance swap. One common type is the gamma swap, where:

$$\text{payoff} = \text{Notional amount x (Gamma – Strike)}$$

$$\text{Gamma} = \frac{252}{n}\sum_{i=1}^{n}\left[\frac{S_i}{S_0}\left(ln\frac{S_i}{S_{i-1}}\right)^2\right] \times 10000$$

Thus, gamma swaps are equivalent to variance swaps where the nominal amount is proportional to the level of the underlying asset. An analytical pricing formula similar to that of variance swap can be obtained (e.g. see Mougeot (2005)).

Gamma swaps have the following characteristics:

- A common strategy, known as dispersion trading, is to sell the index volatility and buy the volatilities of its constituents in order to create a position in correlation. Gamma swap is more suited for this type of trade because unlike the variance swap, re-balancing is not necessary as the weightings would automatically increase when stock prices go up.
- It provides a way to trade the volatility skew (see a later section) – a position of long gamma swap and short variance swap will create an exposure which is long upside variance and short downside variance.
- A cap in the variance swap is no longer necessary as the weighting inherently dampens the downside variance.
- However, given these properties, gamma swaps have never been popular.

Variance futures in the CBOE

With the rising demand for VIX related products, CBOE launched a variance futures contract on the S&P 500 index in December 2012. The contract is very similar to an OTC version of

the variance swap, but with standardized terms such as fixed maturities, trade size per contract (vega notional of US$1000 per contract), and margin requirements. However, the contract does not seem to be accepted by the market, and attract very little interest from participants. A newly re-designed contract would start trading in September 2024, based on the annualized realized variance of the S&P 500 index, which is closely related to the product features of a variance swap.

Applications of variance swaps and modelling issues

If a liquid market exists for the variance swap and there is no need to cross-hedge with the underlying, an accurate model may not be needed. Typically, investment banks are sellers and hedge funds are buyers of these swaps before the financial crisis. Variance swaps are commonly used in taking positions in spread trading of volatilities.

On the other hand, we may use the prices of the liquid variance swaps for the calibration of more sophisticated derivative models, such as models involving jumps in the underlying, so that we can price other exotic products related to volatility. A good model is required if the volatility products need to be hedged. Examples of these products include options on variance (although they are not popular), volatility derivatives, and the correct pricing of the embedded option in the variance swap.

In certain pricing applications, the forward variance of an underlying is needed. Bossu (2014) shows that two variance swaps with maturities $T_1 < T_2$ may be combined to capture the forward variance between T_1 and T_2, which is defined as:

$$T_1\sigma^2_{\text{Realized}}(0,T_1) + (T_2 - T_1)\sigma^2_{\text{Realized}}(T_1,T_2) = T_2\sigma^2_{\text{Realized}}(0,T_2)$$

The fair strike of a forward variance swap is:

$$K_{\text{var}}(T_1,T_2) = \sqrt{\frac{T_2}{T_2 - T_1}K^2_{\text{var}}(0,T_2) - \frac{T_1}{T_2 - T_1}K^2_{\text{var}}(0,T_1)}$$

i.e. the hedge is to short $\frac{T_1}{T_2-T_1}exp(-r(T_2 - T_1))$ of the T_1 maturity variance swap, and long $\frac{T_2}{T_2-T_1}$ of the T_2 maturity variance swap, where r is the interest rate between T_1 and T_2.

If we assume the realized variance is tradable, the standard lognormal model can be applied (following the suggestion in Bossu (2014), pp.65-66). However, unlike the constant volatility assumption in a standard Black-Scholes framework, by definition realized variance is a mixture of past variance and future variance:

$$v_t = \frac{t}{T}\sigma^2(0,t) + \frac{T-t}{T}K^2_{var}$$

v_t is the forward price of the variance started at time *0* and ending at time T, and the first and second term represent the historical and the future variance as observed at time t respectively. Hence the volatility of v_t cannot be a constant. One can modify the standard lognormal framework and assume dynamics such as:

$$dv_t/v_t = 2\omega\frac{T-t}{T}dW_t$$

where ω is a volatility of (future) volatility and 2 is the factor to convert variance to volatility. In this set up, the diffusion coefficient decreases to 0 linearly towards maturity. The fair strike of a volatility swap can be derived as:

$$K_{\text{vol}} = K_{var}\, exp\left(-\frac{1}{6}\omega^2 T\right)$$

If volatility swap and variance swap prices are available, we can make use of the above formula to calibrate ω, and the model can then be used to price other volatility derivatives.

8.2.3. Trading the implied volatility surface

8.2.3.1. Volatility skew/smile

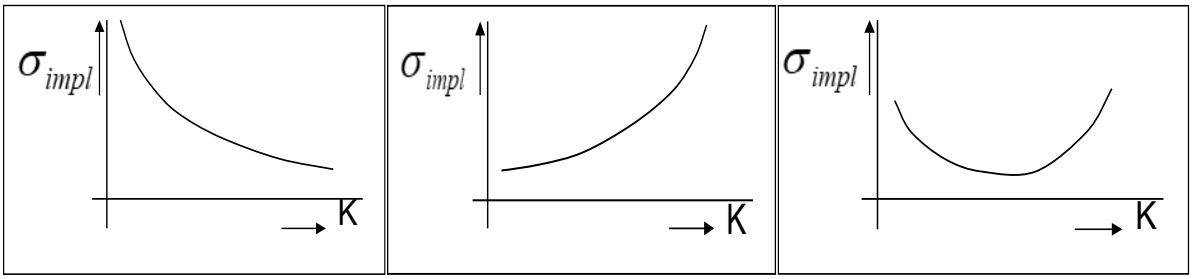
Market data

Representing different volatilities with different maturities is called the **term structure of volatility**. In real markets (especially for stock indices), volatility often exhibits another characteristic, known as **volatility skew** or **smile**. In addition to varying volatilities across time, options at the same maturity but different strikes would be quoted at different volatilities. A typical set of market volatility quotes looks like these:

		Strike	
Maturity	**90**	**100**	**110**
3Mth	47%	44%	41%
6Mth	44.5%	42%	38%
1 Year	39%	37%	35%
2 Years	35%	33%	32%
3 Years	32%	31%	30%
4 Years	30%	29.5%	29%
5 Years	29%	28.5%	28%

Theoretically speaking, this phenomenon is counter-intuitive, because the volatility should describe the movement of the underlying variable (e.g. a particular stock's price); it should have nothing to do with the characteristic of the option (i.e. strike). Also, we cannot generate this kind of result from using the standard formulas with the historical time series. However, the "volatility" has been used as an alternative way to express the price of an option. Rebonato (1999, p.78) explains the phenomenon like this: smile implied volatility is "a wrong number to put in the wrong formula to obtain the right price."

In different markets, the typical shapes of the skew can be very different. Since the crash in 1987, equity markets show a strong negative skew, but Gold often shows a positive skew, whereas currencies can have symmetric volatilities, as shown in the figures below (Dupire (2011)).



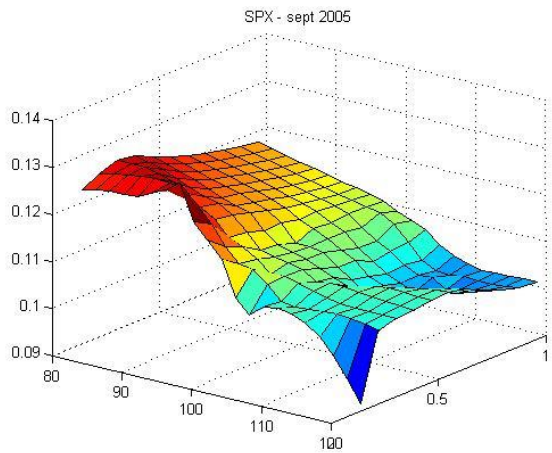
The most common explanations of the smile phenomenon include:

- supply and demand of options at different strikes – more people demand downside puts than upside calls, and there are many demands for zero cost collars (long stock, long put, sell call combination).
- the underlying distribution is not lognormal; it could have fat tails or jumps.
- the volatility of the underlying is stochastic.

Recovering the theoretical volatility skew from historical price data

For most stocks, historical price data are readily available, but the stock price distribution can normally only generate a constant volatility. Dupire (2011) proposes the following procedure in generating the "theoretical skew" from historical price data alone:

1. Price and delta hedge an option, strike K_i, maturity T_i, at volatility σ_i.
2. Compute the P&L in the final account.
3. Solve for σ_i such that the P&L = 0 (c.f. section 6.1.4.2).
4. Repeat the above procedure for different K_i and T_i.



The above diagram shows the volatility skew for S&P 500 in year 2005. While it is theoretically possible, real data for most stocks do not give rise to recognizable skew shapes.

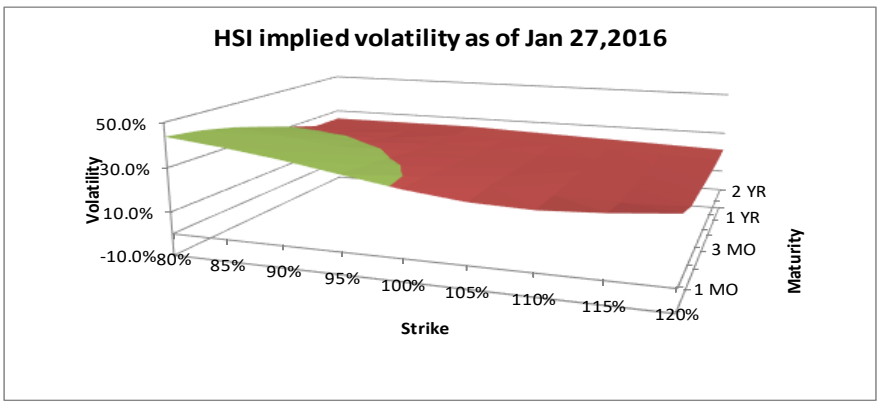
8.2.3.2. Representing the volatility surface

CALL				Feb-2022			PUT			
OI	Volume	IV	Bid/Ask	Last	Strike	Last	Bid/Ask	IV	Volume	OI
417	6	25.2%	255 / -	1,035	**24,000**	159	91 / 176	24.7%	800	817
426	14	24.8%	229 / -	861	**24,200**	202	107 / 250	24.6%	567	550
950	89	23.5%	580 / 777	728	**24,400**	256	200 / 300	23.9%	644	639
737	170	23.1%	400 / 650	572	**24,600**	308	305 / 357	22.5%	474	382
1,108	461	22.7%	360 / 486	455	**24,800**	390	167 / 550	22.5%	483	363
1,110	520	22.1%	258 / 400	370	**25,000**	480	- / 862	22.3%	251	222
1,414	822	21.0%	208 / 371	270	**25,200**	616	599 / -	19.1%	17	83
1,006	638	21.4%	130 / 213	200	**25,400**	802	- / -	21.4%	12	71
988	474	20.6%	133 / 207	145	**25,600**	864	- / -	21.1%	1	4
749	529	20.3%	75 / 128	102	**25,800**	1,058	- / -	20.9%	1	16
1,160	277	20.6%	60 / 80	70	**26,000**	-	- / -	-	-	3

The table above shows the listed options of HSI as of Feb 11, 2022 (source: hkex.com.hk). [Implied volatility (IV) calculation assumption: interest rate 0.15%p.a., dividend yield 4.24%p.a.; OI: Open Interest]

An example volatility surface from Jan 27, 2016 is shown below:

Term	80%	85%	90%	95%	100%	105%	110%	115%	120%
1 MO	43.8%	41.0%	37.5%	33.3%	29.2%	26.2%	25.4%	26.7%	29.0%
2 MO	41.9%	38.6%	35.2%	31.7%	28.7%	26.2%	24.5%	24.1%	24.8%
3 MO	38.7%	36.0%	33.0%	29.7%	27.3%	25.4%	24.2%	23.7%	23.8%
6 MO	34.3%	32.3%	29.9%	27.4%	25.8%	24.4%	23.6%	23.1%	22.7%
1 YR	29.8%	28.5%	27.1%	26.1%	24.8%	23.4%	22.7%	22.3%	21.8%
1.5 YR	25.5%	25.5%	25.5%	25.5%	24.6%	23.8%	23.0%	22.1%	21.3%
2 YR	25.5%	25.5%	25.5%	25.5%	24.6%	23.8%	23.0%	22.1%	21.3%



Once these inputs are obtained, trading opportunities may be identified by modelling the volatility surface:

- How should we treat "some" market option prices as inputs, and take advantage of the mis-pricing of particular points in the volatility curve?
- Can we anticipate the movement in the volatility surface, and make a profit from the movement?
- How can we hedge option positions effectively when underlying conditions change?

Practically, it is desirable to have a smooth and well-behaved implied volatility surface rather than having isolated points. Sometimes it is necessary to have the ability to quote prices on strike/expiry which are not liquid, and a functional representation is appropriate. Constructing a surface would usually be based on one of these methods:

- Special interpolation and extrapolation methods from the existing surface;
- Parametric or non-parametric representations of the surface;
- Full scale derivative pricing models, such as stochastic volatility models (Heston, SABR), levy process and/or jump diffusion models.

Examples of simple parametric schemes

In the first example, at each maturity, we can simply define the skew as two sets of parabolic curves, divided at the ATM forward point, subject to cut off points at either end. For example,

$$v(x) = v_0 + sC + LC^2, x \leq C$$
$$v(x) = v_0 + sx + Lx^2, C < x < 0$$
$$v(x) = v_0 + sx + Rx^2, 0 \leq x < D$$
$$v(x) = v_0 + sD + RD^2, D \leq x$$

K is the strike, F is the forward, $x = \ln(K/F)$, C is the lower cut-off point ($C<0$), D is the higher cut-off point ($0<D$), s is the slope, L and R are the left and right curvature, $v(x)$ is the volatility at x, v_0 is the volatility at strike equal to ATM forward.

In the second example, a popular rule of thumb expressing the relationship between the different points in the volatility surface is the square root of time rule (c.f. Natenberg (1994)). It is assumed that the volatility smile is proportional to $\ln(K/F)/\mathrm{sqrt}(T)$ where K is the strike, F is the ATM forward, and T is the time to maturity.

Knowing the volatility smile for options that mature at one particular time T^*, together with the ATM volatilities for other maturities, the specification of the whole volatility surface can be extracted. Assume that at time T^*, the ATM forward is F^*, and $\sigma(F^*,T^*)$ is the volatility at strike F^*; also, the volatility skew is known (i.e. volatilities at different strikes are given). At time T, the ATM forward is F, and the given volatility is $\sigma(F,T)$. We want to find the volatility at any arbitrary strike K.

First, we can calculate the strike level K^* by assuming:

$$\frac{ln(K/F)}{\sqrt{T}} = \frac{ln(K*/F*)}{\sqrt{T*}}$$

The skew can simply be derived from:

$$\sigma(K,T) - \sigma(F,T) = \sigma(K*,T*) - \sigma(F*,T*)$$

Arbitrage in constructed volatility surfaces

The above examples are ad-hoc interpolation rules that may lead to incoherent results. In particular, Homescu (2011) suggests different possible kinds of internally inconsistent arbitrage. Let c and p be the call and put prices. In a call or put spread arbitrage:

$$\frac{\partial c}{\partial K} \leq 0, \frac{\partial p}{\partial K} \geq 0$$

In a butterfly spread arbitrage:

$$\frac{\partial^2 c}{\partial K^2} \geq 0, \frac{\partial^2 p}{\partial K^2} \geq 0$$

In a calendar spread arbitrage:

$$\frac{\partial c}{\partial T} \geq 0, \frac{\partial p}{\partial T} \geq 0$$

Let's look at a butterfly arbitrage example (negative price with positive payoff):

	+1 call K_1 K_1=96	**–2 calls K_2 K_2=98**	**+1 call K_3 K_3=100**	**Price**
Case 1	σ = 27%	σ = 26.5%	σ = 25.5%	0.06193
Case 2	σ = 27%	σ = 26.5%	σ = 25%	–0.01884

In case 1, a butterfly spread has a positive price. However, if a parametric scheme returns the volatilities as in case 2, an arbitrage results, as the butterfly spread has a negative price. A similar example of a calendar arbitrage is shown below (longer maturity option is too cheap):

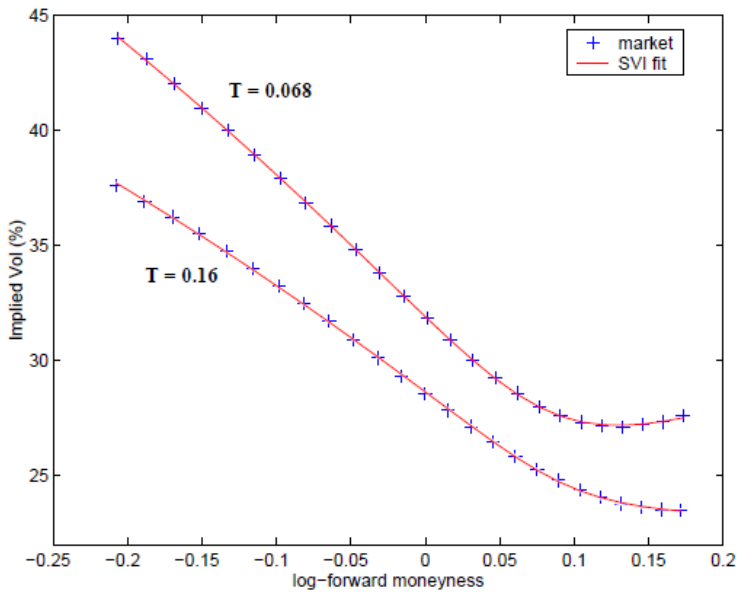
Strike = 110	**–1 call T_1 T_1=30 days**	**+1call T_2 T_2=33 days**	**Price**
Case 1	σ = 15%	σ = 14.5%	0.00221
Case 2	σ = 15%	σ = 14.0%	–0.00309

SVI parametrization

A more sophisticated interpolation scheme, known as stochastic volatility inspired (SVI) parametrization, was first suggested by Gatheral (2004). Each time slice of the implied volatility surface is fitted separately, where the relationship between the implied variance and the function $\ln(K/F)$ is in the form:

$$\sigma^2 = a + b\left(\rho(k-m) + \sqrt{(k-m)^2 + s^2}\right), k = ln(K/F)$$

Additional constraints are imposed on the coefficients so as to ensure no arbitrage. However, the coefficients are not particularly intuitive. Roughly speaking, a controls the overall level of variance, m corresponds to a moneyness shift, ρ is related to the correlation between stock prices and volatility and controls symmetry, s controls the smoothness near the money ($K = F$), and b controls the angle between small and large strikes (c.f. Bossu(2014), p.23).



The above diagram (taken from Zeliade Systems (2009)) shows the calibration of SVI model to the implied smile on the Eurostoxx 50 index on Sep 22, 2008. Practical SVI model fitting may include some constraints on the parameters for optimization, e.g. $a \geq 0, b \geq 0, -1 \leq \rho \leq 1, s > 0$, and a no arbitrage constraint: $b(1+|\rho|)T \leq 4$, where T is the option maturity.

Modelling the underlying asset process

It is also possible to connect the implied volatility surface to the underlying asset process. For example, the local volatility model (Dupire(1994)) has the following form:

$$\frac{dS_t}{S_t} = rdt + \sigma_{local}(t, S_t)dW_t$$

Dupire shows that the function can be obtained by:

$$\sigma_{local}(T,K) = \frac{1}{K}\sqrt{\frac{2\frac{\partial c(T,K)}{\partial T}}{\frac{\partial^2 c(T,K)}{\partial K^2}}}$$

If the implied volatility surface $\sigma^*(K,T)$ is known and smooth, we can solve for the local volatility (Bossu (2014), p.49)

$$\sigma_{local}^2(T,K) = \sigma^{*2} \times \frac{1 + \frac{2T}{\sigma^*}\frac{\partial \sigma^*}{\partial T}}{1 + 2d_1\left(K\frac{\partial \sigma^*}{\partial K}\sqrt{T}\right) + d_1 d_2\left(K\frac{\partial \sigma^*}{\partial K}\sqrt{T}\right)^2 + \left(K\frac{\partial^2 \sigma^*}{\partial K^2}\sqrt{T}\right)\left(K\sigma^*\sqrt{T}\right)}$$

Alternatively, we can go in the opposite direction. Instead of modelling the implied volatility surface directly, we can use a model for the underlying asset process in order to generate the volatility surface. There are two ways to generate a downward sloping implied volatility skew:

- Negative correlation between price and volatility, either through a deterministic dependency (local volatility) or through a negative correlated relationship (stochastic volatility)
- Inclusion of a downward jump component.

The following example is taken from Hull (2009). Assume that the current stock price is $50 and it is known that in 1 month the stock will be either $42 or $58 (possibly some news causing the jump). With a risk-free rate of 12%, the option prices and implied volatilities are calculated as shown here:

Strike($)	42	44	46	48	50	52	54	56	58
Call price($)	8.42	7.37	6.31	5.26	4.21	3.16	2.10	1.05	0.00
Implied volatility	0.0%	58.8%	66.6%	69.5%	69.2%	66.1%	60.0%	49.0%	0.0%

8.2.3.3. Volatility regimes

The above section describes the static properties of volatility surfaces. The next step is to model its dynamic behaviour. It is evident when the underlying spot price moves, the surface will change accordingly. But how? Knowing this would affect how we should delta hedge our option positions, especially in situations when vega hedge may not be possible.

Derman (1999) has proposed three possible regimes: (i) sticky strike; (ii) sticky delta; (iii) local volatility model. We use the following example in the discussion below:

Position:	long European call, strike = 100	
Current spot price S:	100	
At-the-money volatility:	26%	
Market volatilities:	Strike	Volatility
	90	28%
	95	27%
	100	26%
	105	25.5%
	110	25%

Sticky strike

In the set up in most of the primitive trading systems, a simple assumption is that the fixed strike volatility is independent of underlying price S. Therefore, at-the-money (ATM) volatility will fall with rising S. For example, if spot price moves from 100 to 105, the original position (100 strike call) would still be priced with a volatility of 26%. For the current ATM call (at strike 105), the volatility is 25.5%. Since we are long the 100 call and the volatility hasn't changed, the delta hedge should work: the option delta calculated with this model is the same as the Black-Scholes delta.

Sticky delta

If we look at the volatility from a historical time series, we may assume the ATM volatility should be independent of the underlying price level. One explanation is that the ATM volatility is supposed to be an estimate for the future cost of replicating the option. Therefore, as S increases, every strike's volatility would need to go up. Using the original volatility skew, we may see that the strike levels have changed, as in the following example:[4]

Market volatilities:	Strike	Volatility
	95	28%
	100	27%
	105	26%
	110	25.5%
	115	25%

In our portfolio of long call + short shares, if spot price moves to 105, the long call position would be more valuable since volatility has moved up from 26% to 27% (strike of the option is 100). To have zero P&L, the short delta should be larger than the Black-Scholes delta.

Local volatility model

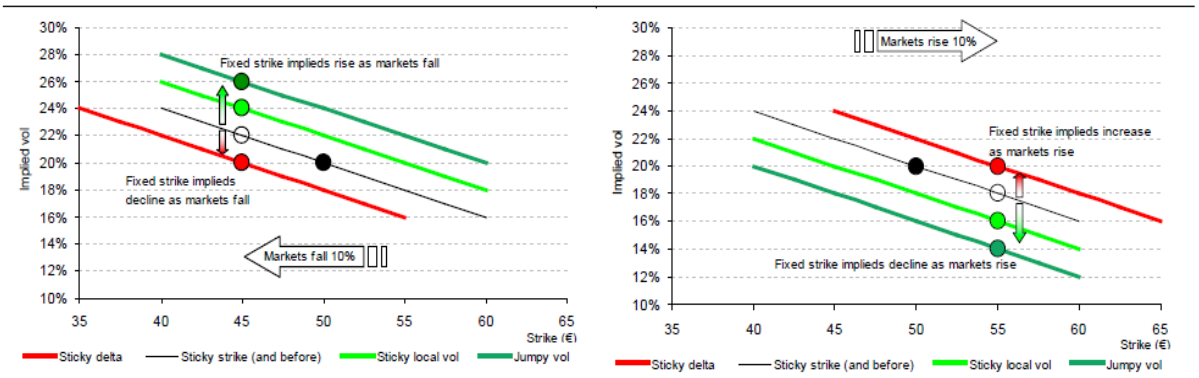
As in our earlier discussions, sticky strike and sticky delta schemes may violate arbitrage relationships (section 8.2.3.2). We could construct an internally consistent model which could match all the skew and maturity data (e.g. local volatility models, as developed in Dupire (1994), Derman and Kani (1994); see section 2.4.2). With this kind of model, we could calculate the shape of the skew when market moves. It turns out that fixed-strike volatility decreases as S increases, e.g. if spot moves to 105, we could have the following:

Market volatilities:	Strike	Volatility
	95	26%
	100	25%
	105	24%
	110	23%
	115	22%

This is the inverse of the sticky delta assumption. Spot price moves up and the volatility of the strike 100 option moves down, making the option position less valuable compared to the Black-Scholes constant volatility assumption. In order to have a correct hedge, the delta of this model would be less than the delta of the Black-Scholes model.

These regimes are summarized as follows:

[4] In a trading system, it may be more convenient to represent the strikes as the moneyness. For example, instead of having the 105 strike call, the volatility level (26%) is stored as representing a 100% ATM strike call.



Source: Bennett and Gil (2012)

Stickiness model	**Fixed-strike Option volatility**	**ATM option volatility**	**delta**
Sticky strike	Independent of index level	Decreases as index level increases	$=\Delta_{BS}$
Sticky delta	Increases as index level increases	Independent of index level	$>\Delta_{BS}$ (adjustment = vega x slope of vol skew)
Local volatility	Decreases as index level increases	Decreases twice as rapidly as index level increases	$<\Delta_{BS}$ (has to be calculated from the model)

Source: Derman (1999)

Concept check

1. Using the volatility term structure at the start of this section, you hold a long position in a call option at strike 100, with volatility = 26%; stock price is currently 100 and you hedge with the Black-Scholes delta (long call + short shares).

<u>Question</u>: When stock price moves to 105, what is the expected volatility of the option at strike 100? What is your expected profit / loss?
<u>Answer</u>:

- Under sticky strike, volatility at strike 100 remains unchanged, hence there is no expected P&L.
- Under sticky delta, volatility at strike 100 may change to 27%, and the long call + hedge position will make money.
- Under local volatility, volatility at strike 100 may change to 25%, and the long call + hedge position will lose money.

2. Hedging in the sticky delta environment

Spot	Volatility	Price	Delta	Vega
100	26%	10.3434	0.5517	0.3956
105	27%	13.6854	0.6238	
P&L change		3.3420		

- Position in a call option, Strike 100, rate = 0%, maturity 1 year.
- When spot price = 100, volatility is 26%. When spot price moves to 105, under the sticky delta environment, volatility of an option with strike 100 is 27%, and the P&L of the un-hedged position is 3.3420.
- If we hedge the initial position with the B-S delta, the P&L from the delta hedge is: 0.5517x5 = 2.7585.
- If we hedge with the adjusted delta, delta = B-S delta + vega x slope of vol skew = 0.5517 + 0.3956 x 0.2 = 0.6308; the P&L is: 0.6308 x5 = 3.1541
- The hedge with the adjusted delta is not perfect as the P&L does not match the shift in P&L of the call position (=3.3420), mainly because the gamma of the position is not taken into account.

It is evident that the delta adjustment is only applicable if no vega hedge is applied to the original position. In this case, the change in volatility is due to the shift in volatility surface when spot price moves, but volatility changes can also be affected by other factors. Sometimes it is not easy to isolate the particular reasons leading to volatility changes. However, the method is still useful, because for some illiquid underlying assets, no instrument is available for hedging the vega, and a trader has to use the delta hedge as a proxy for vega hedge.

Which regime works?

In the study by Derman (1999), he concludes that different market conditions can favor different regimes. One should use:

- sticky strike if the view is that the index is range-bound without a significant change in current realized volatility.
- sticky delta if the index is trending (i.e. undergoing a significant change in level), without a significant change in realized volatility. Skew shape reflects the risk premium for hedging errors and jumps in the market, which is assumed to be unchanged.
- local volatility if jumps (especially down jumps) are likely.

Empirically speaking, all three regimes seem to be working at one time or another and no single regime dominates. For example, Crepey (2002) studied FTSE data and concluded that it is difficult to generalize the result as volatility is stochastic. Daglish, Hull and Suo (2006) used 47 months of OTC data on S&P 500 and suggested that the sticky delta rule outperforms the sticky strike rule.

8.2.3.4. Examples of trading strategies under the various regimes

Arbitrage in the sticky strike model

It can be shown that the sticky strike regime is not arbitrage free. Recall that the P&L of a delta-hedged position in a period Δt is given by:

$$\Delta P = \frac{1}{2}\Gamma S^2(\sigma^2 - \Sigma^2)\Delta t$$

where σ is the realized volatility and Σ is the implied volatility. Assume there are two call options, $C_1 = C_{K_1,T_1}$, $C_2 = C_{K_2,T_1}$, $\Sigma_1 > \Sigma_2$. The P&L of the delta hedge position for each option is:

$$\Delta PL(C_i) = \frac{1}{2}\Gamma_i S^2(\sigma^2 - \Sigma_i^2)\Delta t,\ i = 1,2$$

We can form a portfolio $\Gamma_1 C_2 - \Gamma_2 C_1$, and the P&L is thus:

$$\Delta PL(\Gamma_1 C_2 - \Gamma_2 C_1) = \frac{\Gamma_1\Gamma_2}{2} S^2(\Sigma_1^2 - \Sigma_2^2)\Delta t > 0$$

By definition, this portfolio is delta neutral, and the gamma is also 0. However, it gives rise to a positive P&L in a period Δt , which means that its theta is positive! [Normally, unless the position has negative convexity, it cannot give rise to a positive time decay].

Short skew trades in the sticky delta model

A profitable skew trade can be constructed in a quiet and trending market (with no jumps), when the sticky delta regime is usually prevalent. We can hold the following risk reversal portfolio PF: short put at strike K_1, long call at strike K_2, with $K_1 < S_0 < K_2$, i.e. $PF = C_{K2} - P_{K1}$. By construction, we want to keep the delta as 0.

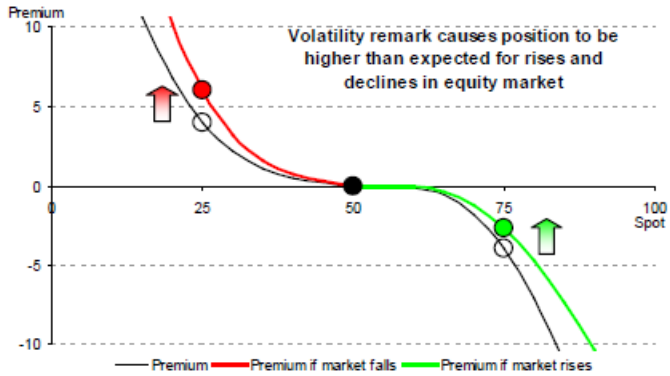
Now there are two possible scenarios:

- If the underlying stock price S increases, both σ_1 and σ_2 will increase. But since vega K_2 > vega K_1, the value of PF will increase because its net vega is positive.
- On the other hand, if the underlying stock price S decreases, both σ_1 and σ_2 will decrease. However, vega K_2 < vega K_1, and the value of PF will still increase.

Also, the portfolio PF has a positive time decay, as theta of C_{K2} is smaller than theta of P_{K1} (due to lower volatility). Again, we have portfolio which is long gamma and have a positive theta!

Long skew trades in normal market conditions

A simple risk reversal trade may benefit from the volatility skew. Assume that the correlation between spot price and implied volatility is negative, which can frequently be observed. The investor could have a combined position in long OTM put and short OTM call.



Source: Bennett and Gil (2012)

When markets fall, the primary driver of the value of the position is the put (nearer to ATM), and the put value will increase due to the rise in implied volatility (due to negative correlation with spot). Conversely, when markets go up, the call will be less negative because of falling volatility. Unlike the trades described in the previous section, this position has a negative theta, similar to long positions in vanilla options.

8.2.3.5. Concluding remarks

Ideally, we want to have a model which is arbitrage free and can generate a realistic forecast of forward volatility (c.f. section 8.2.2.3). The last point is especially important if we need to price and risk manage certain exotic options, and the presence of arbitrage may cause problems in model implementation. However, it should be noted that arbitrage constraints for the volatility surface are mathematical constructs which are rarely realized in practice.

Modelling the volatility surface is more of an art than science. In realistic trading conditions, it is difficult to forecast the correct volatility regime. Even the number of inputs are uncertain, as some of the market prices may not truly reflect the trading conditions, and a decision should be made concerning the points that can be treated as inputs. Profitable trades often come from a correct view of the change in volatility surface due to changing market conditions, and market experience is of paramount importance.

Selected references

Bossu, S., E. Strasser, R. Guichard (2005), "*Just what you need to know about variance swaps*," J.P. Morgan research report.

Bouzoubaa, M. and Adel Osseiran (2010), *Exotic Options and Hybrids*, Wiley, Chap 16.

Demeterfi, K., E. Derman, M. Kamal, J. Zou (1999), "*More Than You Ever Wanted To Know About Volatility Swaps*," Goldman Sachs Quantitative Strategies Research Notes.

Derman, E. (1999), "*Regimes of Volatility*," Goldman Sachs Quantitative Strategies Research Notes.

Hull, John (2009), *Option, Futures, and other Derivatives*, 7th edition, parts of Chapters 18 and 21,

Tsay, Ruey (2005), *Analysis of Financial Time Series*, 2nd edition, Chapter 3 (up to section 3.5).

Good reference books on the mathematics of volatility modelling: Jim Gatheral, *The Volatility Surface*, Wiley (2006) and Sebastien Bossu, *Advanced Equity Derivatives*, Wiley (2014).

Interesting articles on volatility forecasting: John Knight and Stephen Satchell eds. (2007), *Forecasting Volatility in the Financial Markets*, 3rd edition, Oxford: Butterworth-Heinemann.

Practical aspects of volatility trading: Colin Bennett, *Trading Volatility* (2014), https://www.trading-volatility.com/Trading-Volatility.pdf; earlier version in Colin Bennett and Miguel A. Gil, *Volatility Trading: Trading Volatility, Correlation, Term Structure and Skew*, Bank Santander (2012).